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INS321202 - BLOCKCHAIN AND CRYPTOCURRENCY

GROUP 8

VERIDOCCHAIN: A BLOCKCHAIN-BASED DECENTRALIZED CREDENTIAL FRAMEWORK

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Table of Contents

List of Figures and Tables

Acknowledgment

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PART I – INTRODUCTION & STARTUP VISION

1. Introduction

1.1. Background of Digital Credentials and Diploma Fraud

In recent years, digital transformation has become an inevitable trend in the education sector on a global scale. The rapid growth of online learning, cross-border education, international joint programs, and global labor mobility has significantly increased the demand for fast, accurate, and trustworthy verification of academic credentials, certificates, and educational records. For employers, educational institutions, and regulatory authorities, the ability to verify academic information plays a critical role in ensuring workforce quality and maintaining the credibility of education systems.

However, current traditional credential verification methods still suffer from fundamental structural limitations. Most verification processes rely on centralized management models, in which academic data is stored and controlled independently by individual educational institutions. When verification is required, stakeholders must go through multiple manual steps such as submitting formal requests, exchanging emails, or relying on intermediary organizations. These processes are not only time-consuming but also incur substantial administrative costs, particularly in cross-border or international verification scenarios.

Beyond operational inefficiencies, academic credential fraud has become an increasingly serious challenge for both education systems and labor markets. Advances in printing technologies and digital editing tools have made forged diplomas and certificates more sophisticated and harder to detect. In the absence of an independent and globally verifiable authentication mechanism, employers and regulatory bodies face significant difficulties in distinguishing legitimate credentials from fraudulent ones. This situation undermines trust in academic qualifications and poses risks to recruitment quality and institutional credibility.

Furthermore, in the context of globalization, differences in data standards, educational management systems, and legal frameworks across countries further complicate the recognition and verification of academic credentials. Dependence on localized verification systems or centralized intermediaries reduces flexibility and limits scalability, particularly when credentials need to be validated across national and institutional boundaries.

Given these challenges, there is a growing need for an academic credential verification solution that is decentralized, transparent, minimally dependent on intermediaries, and capable of operating across borders. In response to this need, **VeriDocChain** is proposed as a blockchain-based startup project designed to address credential fraud, optimize verification workflows, and establish a trustworthy digital academic identity platform suited to the requirements of the digital era and global integration.

1.2. Problem Statement and Research Objectives

VeriDocChain is designed as a decentralized credential verification platform that utilizes blockchain as a trusted infrastructure layer to store cryptographic proofs of academic credentials, certificates, and educational records. Instead of storing complete academic data directly on the blockchain, the system records only cryptographic hash values representing the original data. This approach ensures data integrity and tamper resistance while simultaneously minimizing storage and transaction costs.

The VeriDocChain platform is built upon the Self-Sovereign Identity (SSI) model, in which learners are positioned as the central entities with full control over the sharing and usage of their identity data and academic records. This model aligns with modern data protection principles and reduces reliance on centralized data repositories that are vulnerable to security breaches or misuse.

The primary objectives of VeriDocChain include:

- Preventing credential forgery through blockchain-based verification mechanisms, ensuring that any unauthorized modification can be detected;
- Reducing verification time by enabling near-instant validation of credentials for employers and educational institutions;
- Lowering administrative costs by eliminating unnecessary manual verification steps and intermediaries;
- Enhancing data ownership and control for learners, allowing them to proactively share academic information when needed;
- Supporting cross-border academic recognition to meet the demands of the global education and labor markets.

A strategic element in VeriDocChain's development direction is its adherence to global digital identity standards proposed by the World Wide Web Consortium (W3C), specifically Decentralized Identifiers (DID) and Verifiable Credentials (VC). The adoption of these open standards ensures system compatibility and facilitates integration with other SSI platforms worldwide.

Interoperability not only delivers technical benefits but also creates long-term competitive advantages for VeriDocChain. While many existing solutions rely on semi-centralized or proprietary architectures, VeriDocChain aims to establish an open ecosystem that allows educational institutions to participate without being locked into a single service provider. This approach significantly enhances system scalability and increases the likelihood of real-world adoption.

1.3. Scope and Methodology of the Study

This study examines the design, technology, and business feasibility of VeriDocChain as a blockchain-based credential verification startup. The research focuses on current challenges in academic credential management, the development of a decentralized verification framework, and the evaluation of its practical application in real-world contexts.

A qualitative and design-oriented approach is adopted, combining literature review with comparative analysis of existing solutions. Based on these findings, the study proposes a conceptual system architecture, business model, and use cases, followed by a feasibility assessment covering technical, operational, legal, and financial aspects.

2. Startup Vision: VeriDocChain

2.1. Vision and Mission

VeriDocChain aims to become a global digital trust infrastructure for academic credentials and certificates. We envision establishing a new standard where the recognition of educational qualifications is conducted transparently, securely, and without being restricted by national borders or institutional silos. The long-term objective of the project is to build a universal verification network that serves as the "trust layer" for the global knowledge economy.

The mission of VeriDocChain is to empower learners, educational institutions, employers, and regulatory bodies through a decentralized, user-centric verification ecosystem. We are committed to eradicating credential fraud, minimizing administrative burdens, and strengthening trust in international education systems, thereby facilitating the global mobility of human resources.

2.2. Value Proposition

VeriDocChain generates tangible business value for various stakeholders within the educational and labor ecosystems:

- **For Educational Institutions:** The platform optimizes operational workflows by reducing administrative costs and the time required to handle manual verification requests. Storing cryptographic proofs on a decentralized network protects the prestige and reputation of the institution against diploma forgery.
- **For Employers:** The system provides a near-instant verification mechanism ("instant trust"), allowing for the immediate validation of an applicant's

qualifications without the need for expensive intermediaries. This accelerates recruitment speed, minimizes personnel risks, and enhances the quality of human resource management decisions.

- **For Learners:** VeriDocChain ensures absolute data ownership and global portability of credentials throughout their career. Learners can proactively manage and share their competency profiles anytime and anywhere, significantly increasing their access to the global labor market.
- **For Regulatory Bodies:** The system supports enhanced oversight efficiency and regulatory compliance through transparency and data provenance. VeriDocChain serves as a catalyst for cross-border academic recognition, supporting government strategies for educational integration.

2.3. Competitive Advantages and Market Differentiation

VeriDocChain establishes its competitive position by building an open ecosystem that strictly complies with international W3C standards, including Decentralized Identifiers (DID) and Verifiable Credentials, thereby eliminating the risk of vendor lock-in and ensuring global interoperability. Unlike traditional solutions, the project adopts the Self-Sovereign Identity (SSI) model to grant learners full control over their data, enhancing privacy protection in accordance with regulatory frameworks such as GDPR and eIDAS 2.0, while enabling efficient cryptographic verification without storing sensitive information directly on-chain. In addition, the integration of a transparent governance framework with a sustainable financial model following a Web3 SaaS approach allows VeriDocChain to effectively meet the legal and budgetary requirements of educational institutions and the public sector, providing a strong foundation for stable and long-term growth.

PART II – SYSTEM MODEL & TECHNOLOGY FOUNDATION

3. Overall System Architecture

3.1. System Actors and Ecosystem

The VeriDocChain ecosystem is composed of four primary actors, each assuming a critical role in the decentralized lifecycle of academic credentials:

- **Educational Institution (Issuer):** The institution acts as the authoritative source of truth. It is responsible for generating academic credentials, digitally signing them to ensure authenticity, and registering the corresponding cryptographic hashes on the blockchain ledger.
- **Student / Learner (Holder):** The student serves as the primary owner of their digital identity. They receive signed Verifiable Credentials (VCs) from the institution and store them in a secure personal digital wallet, maintaining full sovereignty and control over when and with whom their records are shared.
- **Employer / Organization (Verifier):** This actor requires proof of a candidate's qualifications. The Verifier validates presented credentials by checking the digital signatures of the Issuer and comparing the credential's hash against the immutable record anchored on the blockchain.
- **Blockchain Network (Trust Layer):** The blockchain provides a decentralized infrastructure for trust. It does not store personal data but rather maintains an immutable registry of document hashes, Decentralized Identifiers (DIDs), and revocation statuses, ensuring transparency across the ecosystem.

3.2. Data Flow and Trust Model

The operational workflow of VeriDocChain establishes a secure "Trust Triangle" through a four-stage logic:

1. **Issuance:** Upon confirming eligibility, the educational institution creates a Verifiable Credential, signs it using a private key, and registers its unique cryptographic hash on the blockchain via a smart contract.
2. **Storage:** The signed credential is sent to the student, who stores it off-chain in their SSI-enabled digital wallet. This ensures that sensitive personal information remains

under the student’s direct control and is not exposed on the public ledger.

3. **Presentation:** When applying for a position or further study, the student generates a Verifiable Presentation (VP) from their wallet. This step can utilize selective disclosure, allowing the student to share only the necessary attributes required for the specific request.
4. **Verification & Trust:** The Verifier validates the presentation by performing DID resolution to retrieve the Issuer's public key. By checking the digital signatures and confirming that the VP's hash matches the original hash on the blockchain, trust is established instantly without the need for manual institutional checks.

This trust model eliminates the need for centralized intermediaries while ensuring integrity, authenticity, and non-repudiation across the credential lifecycle.

3.3. On-chain and Off-chain Storage Strategy

A core principle in the VeriDocChain architecture is the strategic separation between on-chain and off-chain data. Sensitive personal data (e.g., grades, full name, date of birth) is not written directly to the public blockchain. Instead, only the cryptographic hashes of the VC/original document are recorded on the chain.

This separation is a prerequisite for resolving the most serious legal conflict facing blockchain technology: conflict with the EU's General Data Protection Regulation (GDPR). Because sensitive data resides in an off-chain storage system and is encrypted, it can be modified, re-encrypted, or deleted at the individual's request, thus complying with privacy requirements. This strategic difference from blockchain models that store more data helps VeriDocChain achieve legal sustainability. Below is an analysis of how data is separated.

Table 1: Data Separation: On-Chain and Off-Chain in the VeriDocChain Model

Data Type	**Content (Example)**	**Storage Location**	**Primary Purpose**
On-chain data	VC/VP hash, Issuer/Holder DID, Revocation status	Blockchain (Ledger)	Integrity, origin verification, anti-tampering
Off-chain data	Detailed diploma content (Name, Grades, Graduation Date)	Secure/Encrypted external storage system	Privacy compliance, allowing editing/deleting (RoBf compliance)

3.4. Deployment Architecture

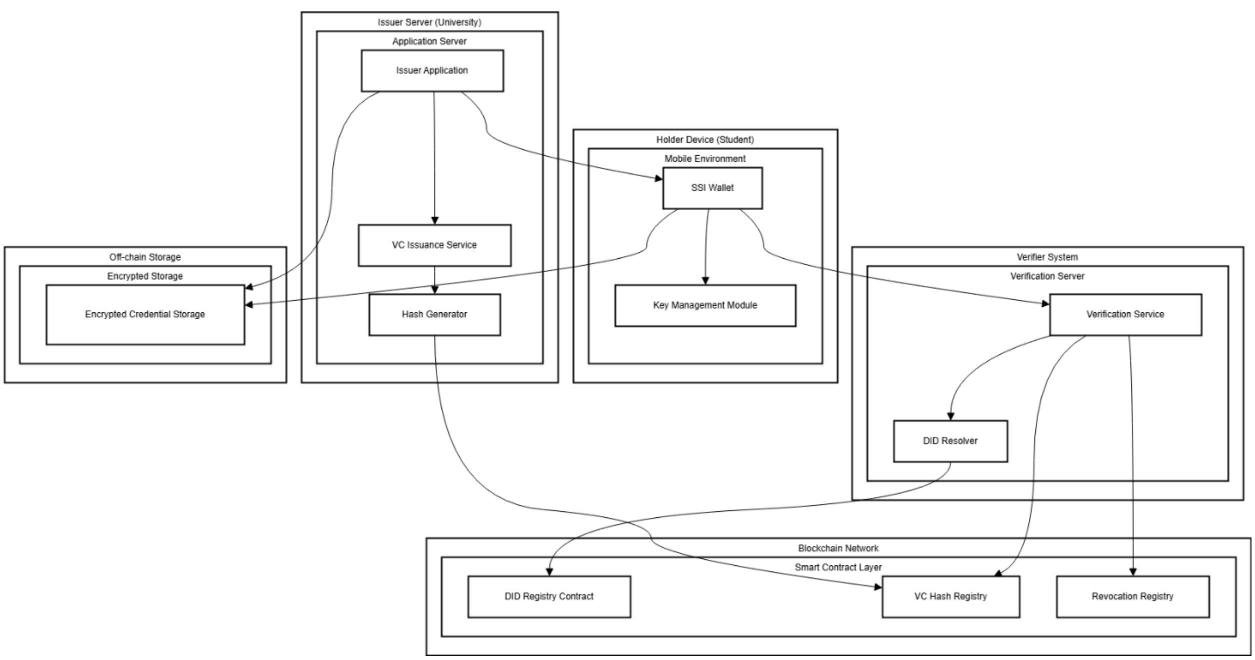

Figure 1: Deployment Diagram of the VeriDocChain system

The VeriDocChain system's Deployment Diagram describes how software components are deployed across different operational environments in real-world scenarios, thereby clarifying the system's distributed deployment architecture. This diagram focuses on mapping key business roles to their respective deployment nodes, rather than detailing functional processing flows.

In the VeriDocChain architecture, the issuer's infrastructure is deployed on dedicated application servers, which operate Verifiable Credentials issuance services, generate hashes, and interact with off-chain storage systems. The certificate holder interacts with the system via mobile devices, using an SSI wallet to store digital certificates and manage cryptographic keys under their direct control. Verifiers deploy independent verification services on their own infrastructure to receive and check Verifiable Presentations as needed.

The blockchain network is deployed as a peer-to-peer distributed infrastructure, providing a common trust layer for the entire system through smart contracts such as the DID Registry, VC Hash Registry, and Revocation Registry. The clear separation of deployment environments ensures the system's security, scalability, and operational feasibility, while also acting as a bridge between the conceptual design model and the actual technical implementation of VeriDocChain.

4. Core Technologies and Standards

4.1. Blockchain as a Trust Layer

In the VeriDocChain system, the blockchain layer acts as a decentralized trust foundation, replacing traditional centralized verification authorities. Rather than storing sensitive personal data, the blockchain is utilized to maintain cryptographic records, including credential hashes, issuer identifiers, and revocation status. This mechanism ensures immutability and provenance, allowing any third party to independently verify the integrity of a diploma without risk of forgery or tampering.

4.2. Self-Sovereign Identity (SSI) Model

The Self-Sovereign Identity (SSI) model is the core philosophy enabling VeriDocChain to shift data control from centralized institutions to the learners. Through this model, learners are issued a Decentralized Identifier (DID) to manage their academic records in personal digital wallets rather than on institutional servers. Adopting SSI not only strengthens privacy and personal data ownership but also ensures strict compliance with modern data protection standards such as GDPR and eIDAS 2.0 by utilizing off-chain data storage.

4.3. W3C Standards: DID and Verifiable Credentials

To ensure cross-border recognition and validation, VeriDocChain is built upon open standards defined by the World Wide Web Consortium (W3C). Specifically, the Decentralized Identifiers (DID) standard provides a unique and permanent digital identification mechanism that is independent of any single service provider. Simultaneously, the Verifiable Credentials (VC) standard defines a standardized data model for digital diplomas, enabling machines to read and perform cryptographic verification automatically. Compliance with these international standards allows VeriDocChain to achieve high interoperability, facilitating seamless connectivity with global education and labor ecosystems.

4.4. Smart Contracts and Cryptographic Verification

Smart contracts are the components that execute automated verification logic within the VeriDocChain network. These contracts are responsible for managing hash registration, recording issuance timestamps, and updating the revocation status of credentials transparently. Combined with cryptographic techniques such as data hashing and Public Key Infrastructure (PKI), the system establishes an irreversible verification process. Any attempt to modify the original credential data will result in a hash mismatch, allowing verifiers to detect fraud immediately and ensuring non-repudiation in academic transactions. Together, these technologies establish a robust technical foundation for secure, decentralized, and globally interoperable credential verification within the VeriDocChain ecosystem.

5. Integration with TrustKeys Platform

In the overall architecture of VeriDocChain, trust is established not only through the data integrity of the blockchain ledger but also depends significantly on the verified identity of the entities behind the transactions. Integrating with the TrustKeys platform acts as a federated trust infrastructure layer, transforming pure technical verifications into an identity ecosystem with legal and operational validity.

5.1. Role of TrustKeys in the Ecosystem

TrustKeys functions as a shared trust infrastructure layer, extending VeriDocChain's trust boundaries beyond the scope of individual educational institutions. While VeriDocChain focuses on the credential issuance and verification processes at the application layer, TrustKeys serves as a "Blockchain Super App" providing digital identity modules, secure key management, and a cross-domain identity trust network. By integrating TrustKeys' "One-Stop" modules, VeriDocChain can standardize the authority of participating organizations, ensuring that every entity acting as an "Issuer" is strictly identity-verified through Public Key Infrastructure (PKI) before joining the network.

5.2. Trust Establishment and Interoperability

The trust establishment mechanism between VeriDocChain and TrustKeys is based on the utilization of "Trust Anchors" and a strictly managed whitelisting process. When an educational institution joins the system, its Decentralized Identifier (DID) and public key are verified and anchored within the TrustKeys trust network.

This architecture creates robust interoperability: verifiers from external systems can query and validate the issuer's identity through the TrustKeys Hub without needing to establish bilateral agreements or custom technical integrations for each individual school. Using TrustKeys as a trust coordination center helps standardize the onboarding process, minimizes trust-establishment costs, and creates a "chain of trust" that spans across diverse identity and credential platforms.

5.3. Strategic Value of TrustKeys Integration

From a strategic perspective, the integration with TrustKeys significantly strengthens VeriDocChain’s market position by accelerating ecosystem adoption and substantially reducing entry barriers for institutional partners.

This integration simultaneously enhances auditability, regulatory compliance, and the system’s long-term competitive capacity. By providing a Public Key Infrastructure (PKI) aligned with international regulatory frameworks such as GDPR and eIDAS, combined with an immutable audit trail for the entire identity verification and credential issuance process, TrustKeys reinforces institutional trust while mitigating both legal and operational risks. Furthermore, leveraging an already established trust network enables VeriDocChain to scale internationally with low integration costs, while creating a sustainable competitive advantage over solutions that focus solely on isolated technical implementations.

This collaboration not only ensures technological sustainability but also lays the foundation for a transparent digital education ecosystem, where trust is established through a tight integration of source code and the real-world identities of human actors.

PART III – USE CASE EVALUATION

6. Core Use Cases of VeriDocChain

6.1. Education Credential Issuance and Verification

Currently, academic credential fraud and slow verification processes cause significant global issues. VeriDocChain provides a blockchain-based SSI solution to ensure diplomas are legitimate, easily verifiable, and support cross-border evaluation.

6.1.1. Identity Onboarding & DID Registration

Before initiating the Core Credential Lifecycle, the VeriDocChain system conducts the Identity Onboarding and DID Registration phase to establish a Self-Sovereign Identity (SSI) infrastructure for the entire ecosystem. This phase defines the processes for creating and registering Decentralized Identifiers (DIDs) for both Issuers and Holders, forming a trusted foundation for all activities throughout the credential lifecycle.

All Issuance, Presentation, Verification, and Revocation processes are cryptographically bound to valid DIDs, thereby ensuring authenticity, traceability, and eliminating dependence on centralized intermediaries.

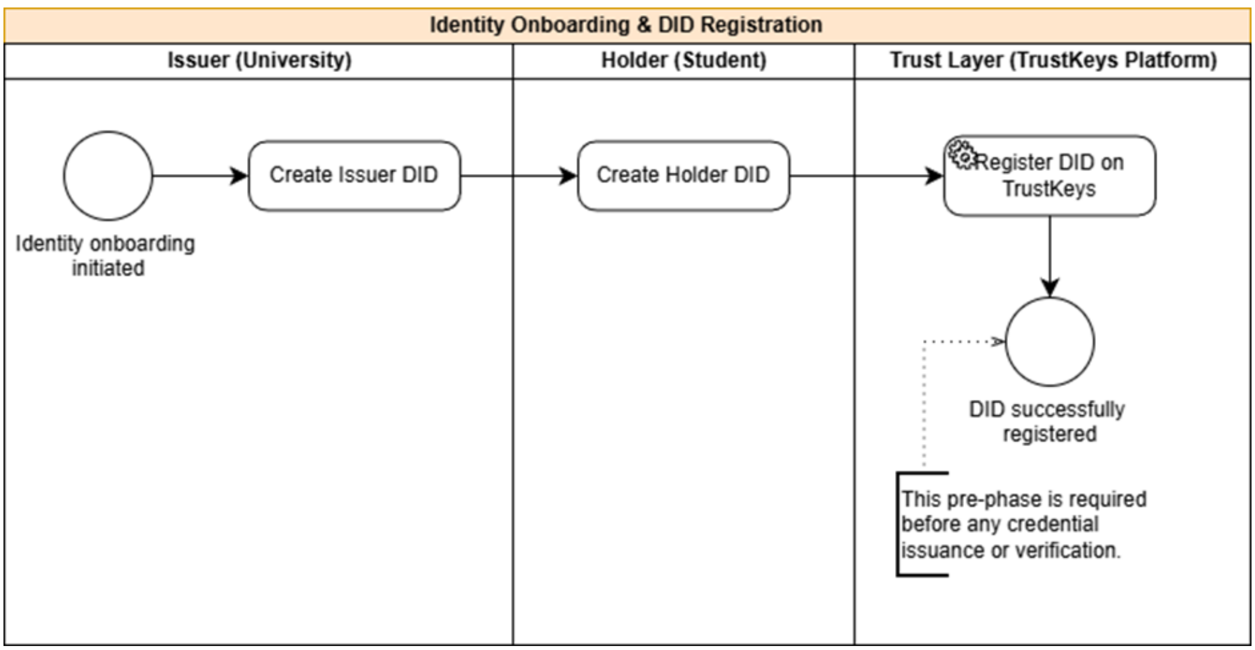

Figure 2: Identity Onboarding & DID Registration Process in the VeriDocChain System

Operational Flow:

1. **Process initiation:** The identity onboarding process is triggered when an Issuer (educational institution) or a Holder (student) joins the VeriDocChain system for the first time.
2. **Issuer DID creation:** The educational institution generates an Issuer DID along with a cryptographic key pair, including a public key for verification and a private

key for signing Verifiable Credentials, thereby establishing the institution's official digital identity within the ecosystem.

3. **Holder DID creation:** The student generates a Holder DID via an SSI wallet, establishing ownership and full control over personal data in accordance with Self-Sovereign Identity principles.
4. **DID registration on the Trust Layer:** The generated DIDs are registered in the DID Registry on the Trust Layer via smart contracts, creating public, immutable, and verifiable identity records for participating entities.
5. **Completion and operational readiness:** Upon successful registration, the identity onboarding phase is completed, and the system proceeds to the Core Credential Lifecycle, beginning with the Issuance process.

6.1.2. Core Credential Lifecycle

The Core Credential Lifecycle describes how VeriDocChain operates after the completion of Identity Onboarding and DID Registration. Based on established and valid digital identities, the system implements a credential management cycle consisting of Issuance, Presentation, Verification, and Revocation, forming a complete lifecycle of a Verifiable Credential. This lifecycle ensures authenticity, traceability, and trustworthiness throughout credential usage.

- **Issuance – Presentation – Verification**

This primary credential flow describes the core operational interactions among the Issuer, Holder, and Verifier, reflecting how the VeriDocChain system operates in real-world credential exchange and verification scenarios.

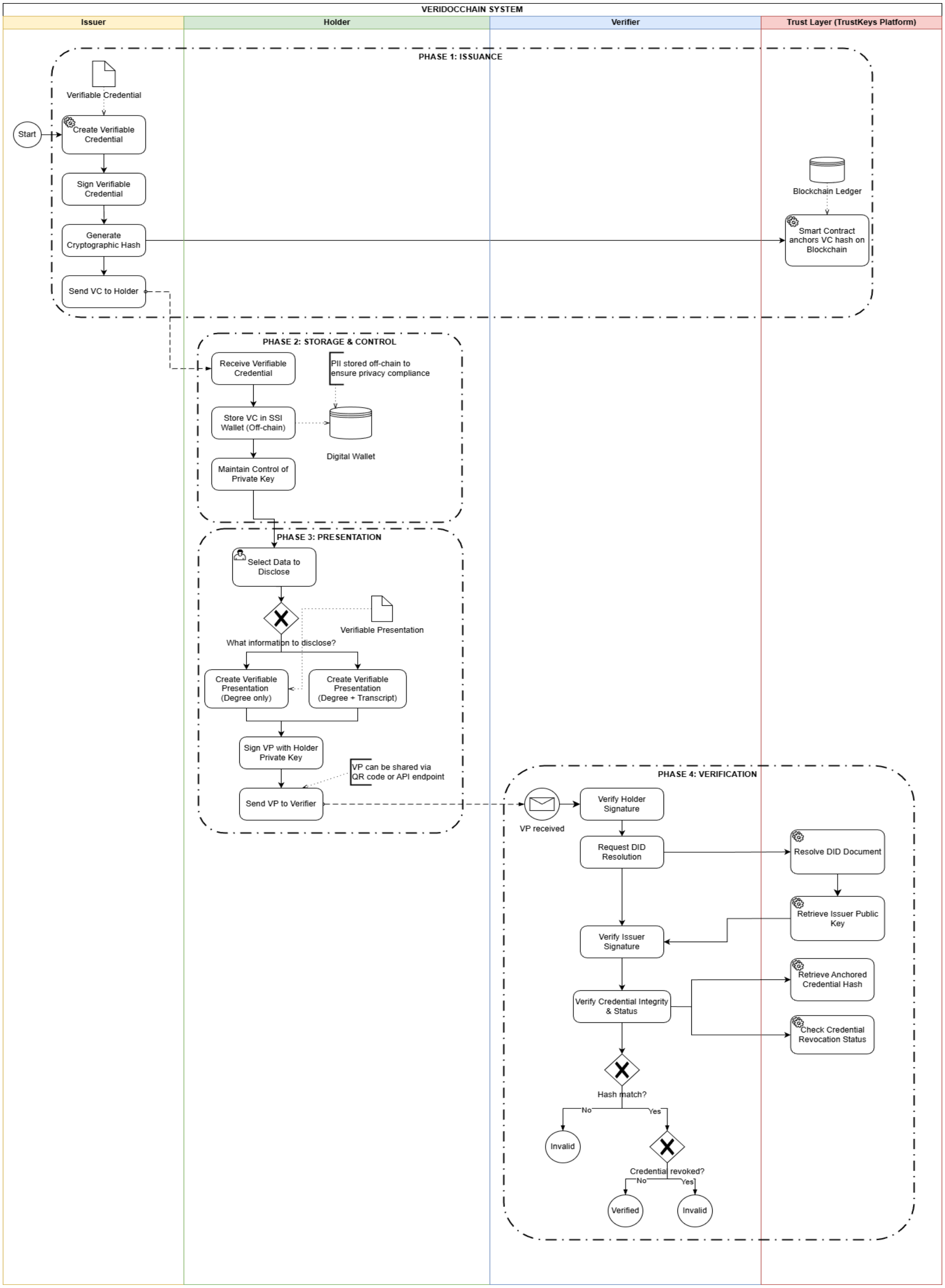

Figure 3: Primary Credential Flow in the VeriDocChain System

Operational Flow:

1. **Phase 1 – Issuance:** The Issuer issues a credential by creating a Verifiable Credential (VC) compliant with W3C standards, digitally signing it using the private key, and computing a cryptographic hash of the VC. This hash is submitted to a smart contract and anchored on the blockchain ledger, creating an immutable record for verification purposes. In parallel, the complete VC is transmitted off-chain to the Holder for storage and use.
2. **Phase 2 – Storage & Control:** The Holder stores the VC in an SSI wallet, where all personally identifiable information (PII) remains entirely off-chain to ensure privacy. The Holder retains full control over the private keys and determines when and how the VC is used.
3. **Phase 3 – Presentation:** When proof is required, the Holder selects the attributes to be disclosed, generates a Verifiable Presentation (VP) from the VC, and applies selective disclosure mechanisms. The VP is signed using the Holder's private key and transmitted to the Verifier via appropriate communication channels.
4. **Phase 4 – Verification:** The Verifier receives the VP, verifies the Holder's signature, and performs DID resolution on the Trust Layer to retrieve the Issuer's public key. The Verifier then validates the Issuer's signature to confirm the credential's origin and queries the original hash and revocation status from the blockchain ledger to assess the integrity and validity of the credential.
5. **Decision Gateway:** If the hash values match and the revocation status is valid, the credential is classified as Verified; otherwise, it is deemed Invalid.

- **Credential Revocation**

The Credential Revocation Process is designed as a complementary mechanism tightly integrated with the Primary Credential Flow, ensuring that credential validity is accurately maintained throughout its lifecycle. While the primary flow focuses on issuance, presentation, and verification, the revocation process provides post-issuance control, enabling the system to invalidate credentials that are no longer valid and to maintain overall ecosystem trust.

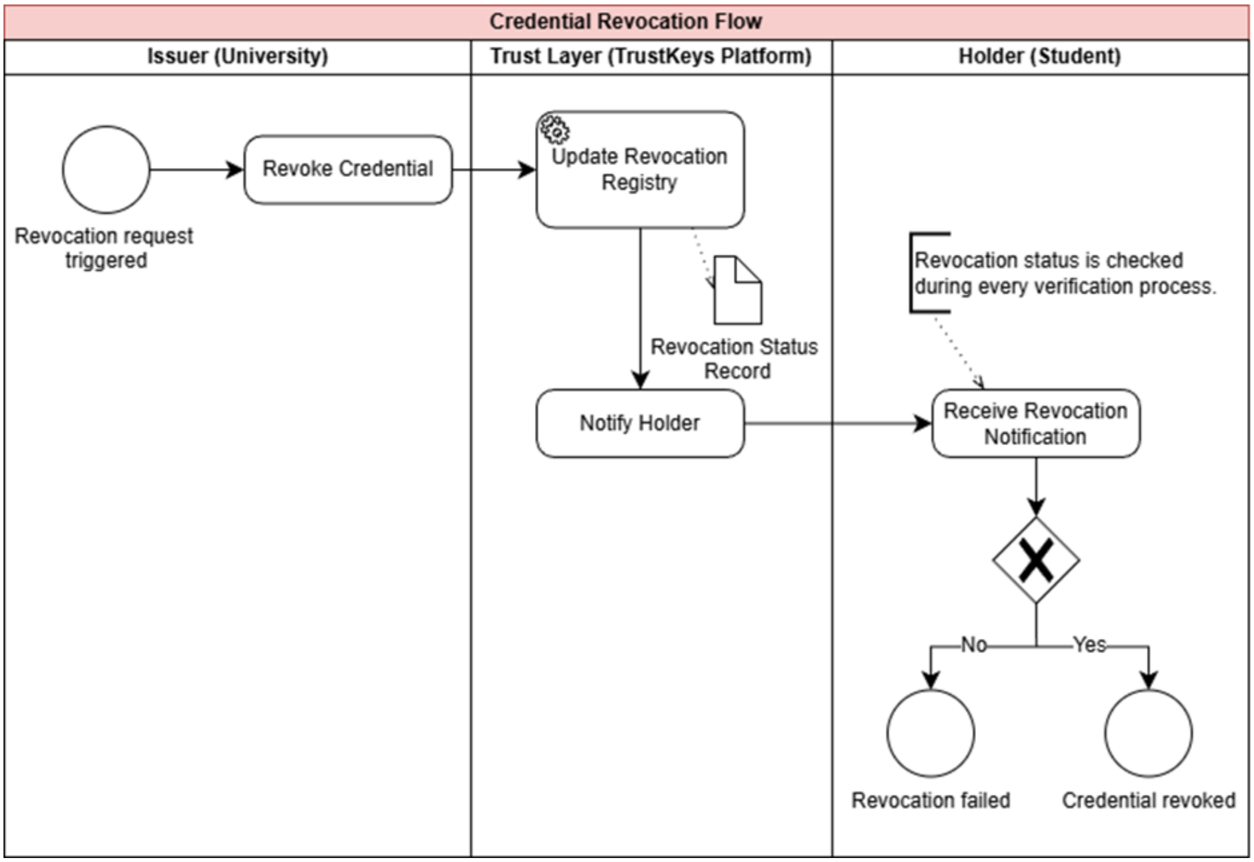

Figure 4: Credential Revocation Process in the VeriDocChain System

Operational Flow:

1. Trigger Event: The revocation process is initiated when the Issuer determines that a credential is no longer valid, due to reasons such as detected fraud, data inaccuracies, or formal revocation requests after issuance.

2. Revocation Request Initiation: The Issuer submits a Revoke Credential request to the VeriDocChain system to formally invalidate the corresponding Verifiable Credential

3. On-chain Revocation Update: Via smart contracts, the Trust Layer updates the Revocation Registry on the blockchain, marking the credential as revoked and recording the revocation event as an immutable on-chain record for future auditing and verification.

4. Holder Notification: The Trust Layer sends a revocation notification to the Holder's SSI wallet, informing them of the change in credential validity status

5. Process Completion: The Holder receives the notification, and the status of the Verifiable Credential within the system is updated to Credential Revoked, completing the revocation process.

6.1.3. Advanced Education Scenarios

In addition to the basic credential issuance, storage, presentation, and verification processes, VeriDocChain supports advanced scenarios in education, including cross-border credential verification and governmental notarization. These processes extend the usability of credentials, enabling legally verifiable checks in multinational and regulatory contexts, while continuing to adhere to the principles of Self-Sovereign Identity (SSI) and blockchain-based trust.

- **Cross-border Verification**

Cross-border Education Credential Verification addresses the challenges of international credential verification—where traditional processes are often slow and complex—by providing a globally interoperable mechanism that allows foreign institutions to directly verify the authenticity of credentials issued in Vietnam in a rapid and transparent manner, without reliance on local notarization intermediaries.

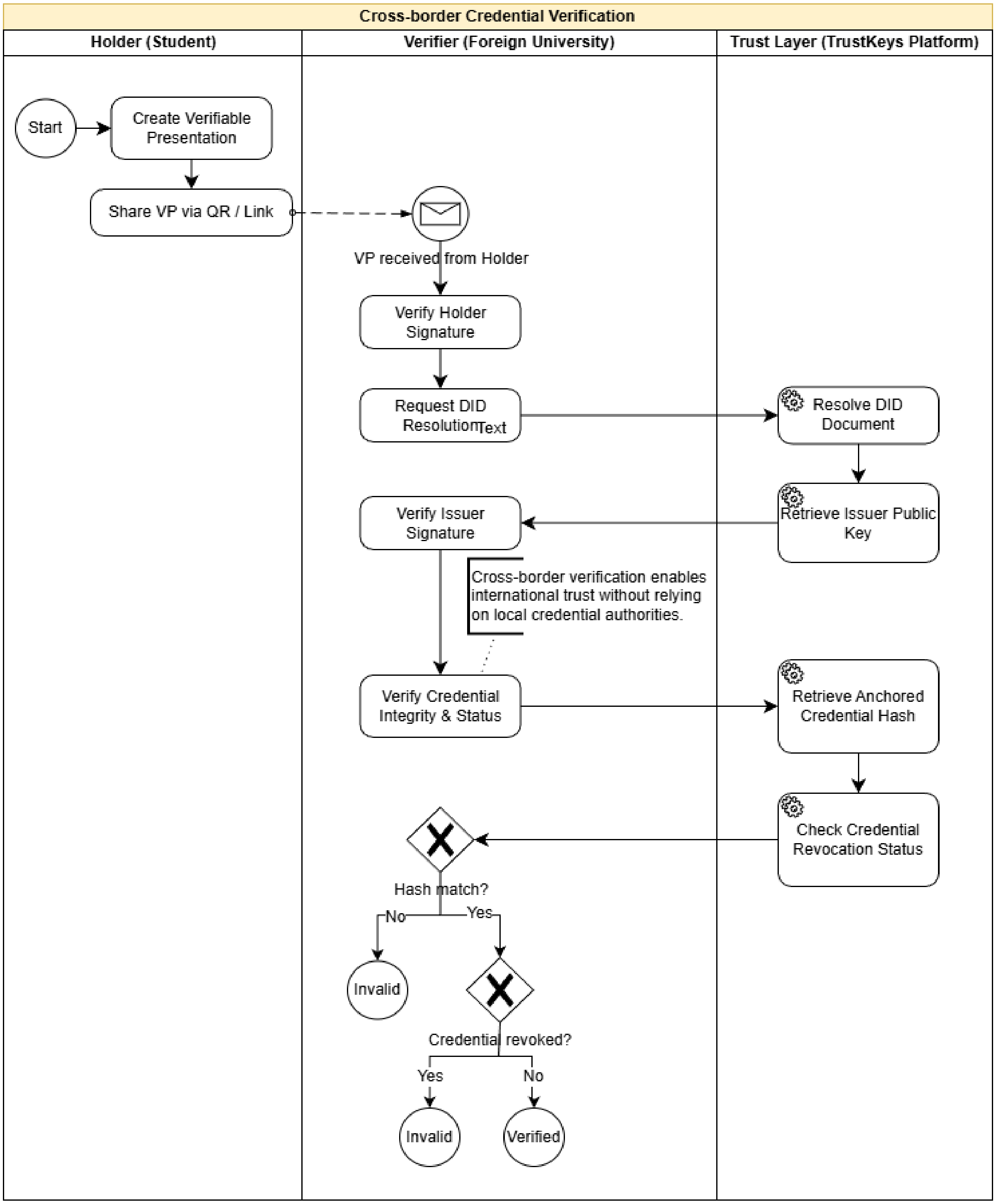

Figure 5: Cross-border Education Credential Verification Process in VeriDocChain

Operational Flow:

1. Verification request initiation: The process begins when a Holder (student) submits credentials to a foreign educational institution (Verifier) during admissions or

academic evaluation.

2. Verifiable Presentation creation and sharing: The Holder uses an SSI wallet to select relevant attributes, generates a Verifiable Presentation (VP) from the original Verifiable Credential, and shares the VP with the Verifier via a QR code or secure link.
3. Source authentication: The Verifier receives the VP and performs DID resolution to query the Trust Layer, retrieves the Issuer's public key, and verifies the Issuer's digital signature to confirm the credential's legitimacy.
4. Integrity and status verification: The Verifier queries the Trust Layer (blockchain) to retrieve the anchored hash and revocation status, comparing them with the VP data to ensure integrity and validity.
5. Cross-border verification completion: Upon successful validation, the credential is marked as Verified, ensuring a secure and verifiable cross-border process.

- **Governmental Notarization & Verification**

Education Credential Notarization and Governmental Verification addresses large-scale credential fraud by providing a trusted verification infrastructure for government authorities, particularly the Ministry of Education and Training. Through the Trust Anchor mechanism on the VeriDocChain platform, only accredited educational institutions are authorized to issue Verifiable Credentials, thereby strengthening regulatory enforcement, preventing fraud, and ensuring nationwide legal validity of educational credentials.

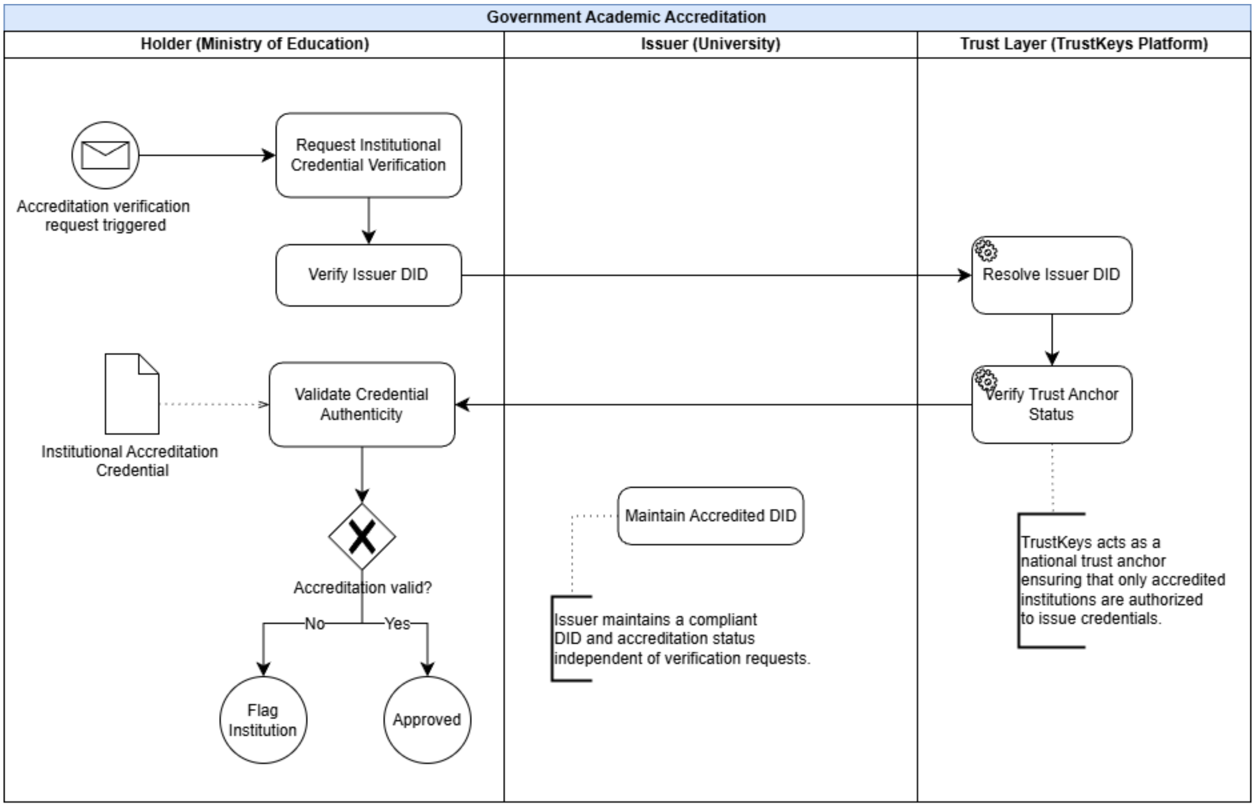

Figure 6: Education Credential Notarization and Governmental Verification Process

Operational Flow:

1. **Institutional verification request initiation:** The process begins when a government authority (Verifier) initiates a verification request for a credential or issuing institution as part of accreditation, inspection, or regulatory activities.
2. **Issuer identity resolution:** The Verifier performs DID resolution to identify the Issuer DID associated with the educational institution and retrieves the institution's identity data and public keys.
3. **Trust Anchor verification on the Trust Layer:** The system queries the Trust Layer (blockchain) to verify the Issuer's Trust Anchor status, determining whether the institution is authorized to issue credentials.
4. **Credential and institutional record validation:**The Verifier validates the credential's authenticity and cross-checks it against registered institutional records to ensure alignment with legal authorization.

5. **Regulatory decision gateway:** If all criteria are satisfied, the institution is marked as *Approved*. If discrepancies or non-compliance are detected, the institution is flagged (*Flag Institution*) for regulatory oversight and enforcement.

These advanced education use cases demonstrate VeriDocChain's applicability for enhanced credential verification processes within the education sector.

6.2. Healthcare Credential Management

In the healthcare sector, verifying the authenticity of professional licenses, specialized certifications, and clinical qualifications is a mission-critical requirement for ensuring patient safety and regulatory compliance. Traditional verification processes are often fragmented and highly vulnerable to sophisticated forms of document fraud, particularly in the context of cross-border medical collaboration. VeriDocChain addresses these challenges by providing a decentralized credential management infrastructure that enables real-time verification and automated revocation of medical credentials throughout the entire professional lifecycle of healthcare practitioners.

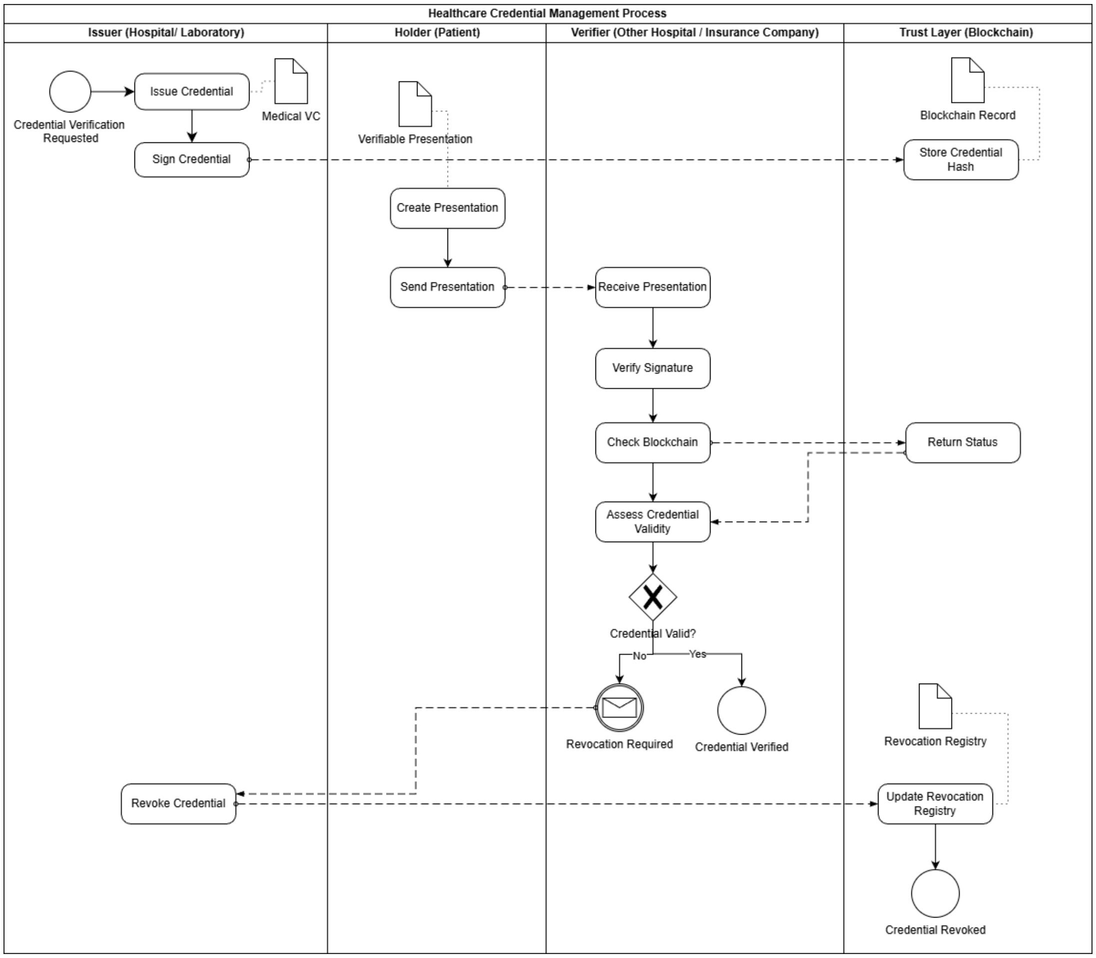

Figure 7: Healthcare Credential Verification and Medical Certification Management Process

Operational Flow:

- Issuance: Authorized healthcare institutions or laboratories (Issuer) generate and digitally sign Medical Verifiable Credentials (VCs). The corresponding cryptographic hashes are anchored on the blockchain.
- Storage: Patients (Holder) receive and securely manage their credentials within personal SSI wallets.
- Presentation: When required, the patient generates a Verifiable Presentation (VP) and submits it to the receiving hospital or insurance provider (Verifier).
- Verification: The Verifier validates the Issuer's signature and checks the credential's current status against blockchain records.

- Revocation: In cases of malpractice, expiration, or non-compliance, revocation status is updated on the ledger and becomes immediately enforceable across the ecosystem.

6.3. Professional Licensing and Certification

Professional workforce management organizations often face the problem of fraudulent credentials and the lack of real-time visibility into the status of professional licenses. VeriDocChain supports transparent lifecycle management of professional certifications (such as engineers, accountants, and lawyers), ensuring instant verifiability and continuous real-time monitoring of license status, thereby enabling organizations to make recruitment decisions based on trustworthy data.

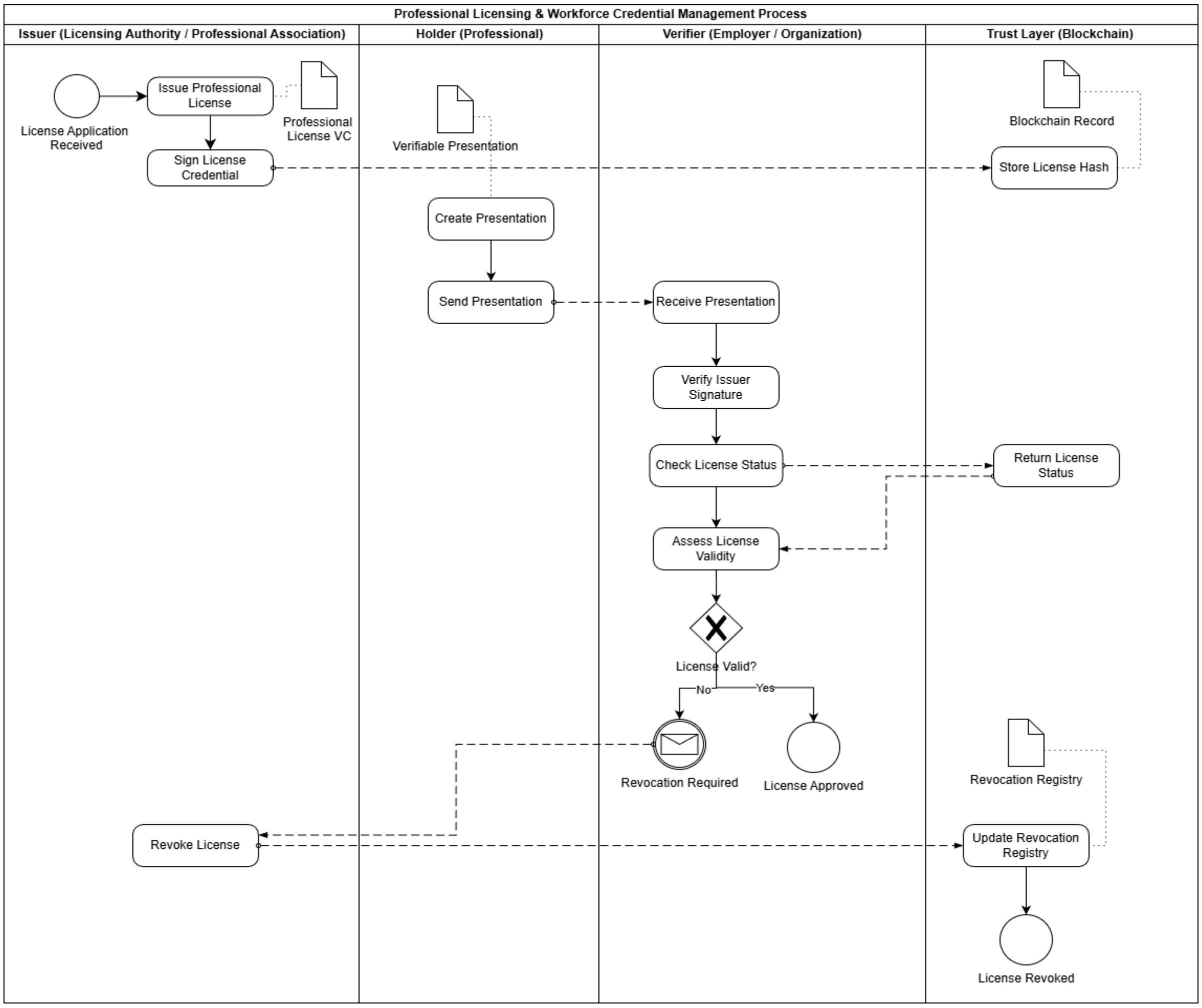

Figure 8: Professional Licensing and Workforce Credential Management Process

Operational Flow:

- Issuance: A professional authority or association (Issuer) evaluates the applicant and issues a digitally signed Professional License VC.
- Blockchain Anchoring: The license hash is registered on the blockchain as a trust reference.
- Presentation: The licensed professional (Holder) submits a VP to employers or partner organizations (Verifier).
- Validation: The Verifier confirms the credential signature and queries the license status on the blockchain to ensure validity.

- Revocation: In cases of professional misconduct or policy violations, the Issuer updates the revocation registry, immediately invalidating the credential across all verification points.

6.4. Supply Chain Certification and Provenance Assurance

The global supply chain is currently facing serious challenges arising from the falsification of certificates of origin and quality labels, resulting in significant compliance costs and the erosion of trust among international partners. VeriDocChain addresses this problem by providing a decentralized infrastructure for managing provenance and compliance certificates in the form of Verifiable Credentials (VCs). This solution enables customs authorities and importing partners to authenticate the legitimacy of goods in real time, ensuring cross-border transparency without relying on paper-based documentation that is highly vulnerable to manipulation.

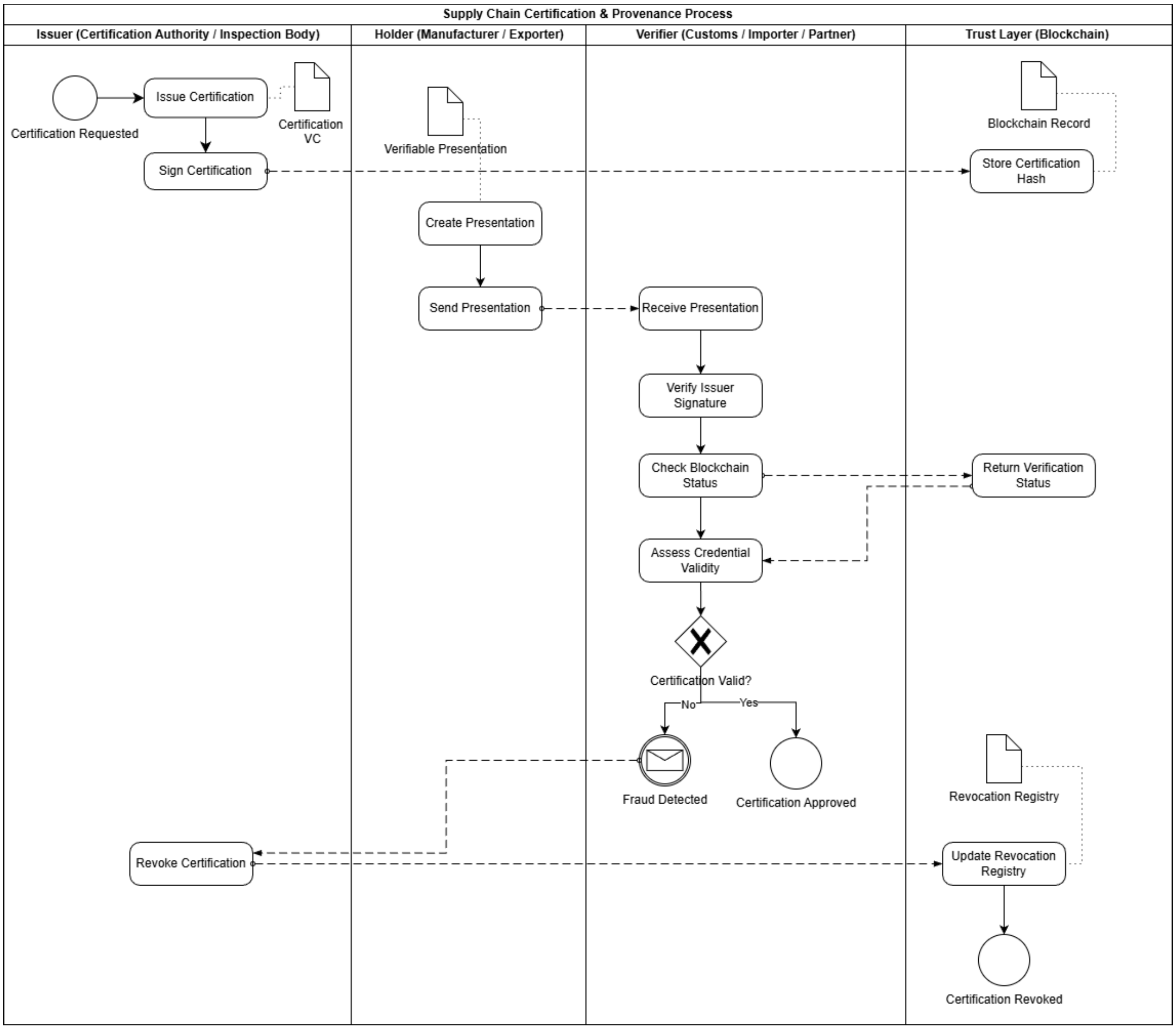

Figure 9: Supply Chain Certification and Provenance Assurance Process

Operational Flow:

- Certification: An accredited inspection authority (Issuer) digitally signs origin and quality certification VCs and anchors their hashes on the blockchain.
- Presentation: Manufacturers or exporters (Holder) present VPs to customs authorities or importing partners (Verifier).
- Verification: The Verifier validates the credential signature and blockchain status to confirm authenticity and compliance.
- Fraud Handling: If fraud is detected, certification revocation is triggered and

recorded on the trust layer, preventing further misuse.

7. Detailed Use Case Analysis

7.1. Stakeholders and Operational Workflow

The effectiveness of VeriDocChain as a cross-sector digital trust infrastructure depends on its ability to coordinate multiple stakeholders within a unified architectural framework. This framework is commonly conceptualized as the "Trust Triangle", in which control over data and verification authority is redistributed among three principal entities: Issuers, Holders, and Verifiers, thereby eliminating reliance on centralized intermediaries. Within the VeriDocChain ecosystem, each participant plays a critical role in maintaining data integrity

Table 2: Stakeholder Mapping and Roles in the VeriDocChain Ecosystem

Stakeholder Group	Role in Ecosystem	Examples
Educational Institutions	Issuers	Universities, vocational colleges, accreditation bodies issuing diplomas and professional certificates.
Healthcare Organizations	Issuers	Hospitals, medical councils, laboratories issuing clinical data, professional licenses, vaccination records
Professional Associations	Issuers	Professional bodies certifying skills and maintaining practitioners' credentials.

Supply Chain Authorities	Issuers	Quality control agencies and customs authorities issuing certificates of origin (C/O), compliance documentation, pharmaceutical traceability records.
Users (Individuals / Enterprises)	Holders	Students, professionals, and companies storing and presenting digital credentials
Employers / Customs Agencies	Verifiers	Organizations consuming credentials for hiring decisions or customs clearance
Government Agencies	Root of Trust & Regulators	Ministries providing digital identities, land ownership records, and legal frameworks
TrustKeys / Blockchain	System Infrastructure	TKBlockchain, API, ví SSI

General Operational Lifecycle of the Ecosystem

Across all application domains—including education, healthcare, professional licensing, and supply chain—VeriDocChain is organized around a unified credential governance lifecycle, serving as the operational backbone of the entire ecosystem.

This lifecycle does not describe the detailed business workflows of individual use cases, but instead establishes a standardized governance model in which all stakeholder interactions follow the same verification and control structure. The framework consists of five core stages:

Issuance → Storage → Presentation → Verification → Revocation

Each stage in the operational lifecycle represents a set of core functional responsibilities of the ecosystem, including ensuring data integrity, protecting the information control rights of the credential holder, maintaining auditability and compliance with legal requirements, and establishing a digital trust infrastructure that operates without intermediaries among participating organizations.

7.2. Value Creation and Benefits

- **Education:**

  In the education sector, VeriDocChain eliminates diploma fraud through immutable cryptographic proof mechanisms. By anchoring credential hashes on the blockchain, the system enables employers and educational institutions worldwide to verify certificates instantly without complex manual verification procedures. This solution reduces administrative costs and verification time by over **90%**, while compliance with international standards promotes academic mobility and global recognition of qualifications.

- **Healthcare:**

  In healthcare, VeriDocChain plays a crucial role in preventing unlicensed medical practice, safeguarding patient safety, and improving service quality. The system allows real-time verification of professional licenses, training records, and practitioner credentials. It enhances interoperability of electronic health records (EHR) while ensuring strict compliance with sensitive data protection regulations. It is estimated that blockchain adoption could save the global healthcare industry up to **USD 100 billion**

annually by 2025 through reductions in data breach risks and IT operational costs. Additionally, the system supports cold-chain pharmaceutical supply management, ensuring vaccine and biomedical product integrity from production to patient delivery.

- **Professional Licensing and Workforce Management:**

In workforce governance, VeriDocChain functions as an "immune system" against legal risks associated with fraudulent or expired certifications. The system automatically monitors credential validity and triggers renewal workflows, reducing regulatory exposure for enterprises. Instant verification of skills and experience accelerates recruitment processes, enabling HR departments to cut **25–50%** of operational and compliance costs while creating a highly adaptive and mobile workforce.

- **Supply Chain and Provenance:**

Within supply chains, VeriDocChain demonstrates its value through counterfeit prevention and transparency enhancement. By assigning unique digital identities to products, the system reduces counterfeit infiltration by up to **75%** across logistics networks. Full digitization of shipping records and certificates of origin improves customs clearance efficiency by approximately **30%**, significantly lowering paperwork and administrative overhead while strengthening cross-border trade trust and sustainability compliance.

- **Government and Public Administration:**

For governments, VeriDocChain supports the construction of reliable digital identity systems for citizens and enterprises, forming the foundation for secure and efficient public service delivery. Blockchain's immutable storage strengthens transparency in critical domains such as land management, taxation, and public procurement, while providing tamper-proof audit trails that enhance regulatory oversight and fraud prevention.

7.3. Feasibility and Adoption Challenges

The large-scale deployment of VeriDocChain requires a multidimensional assessment of technical infrastructure, existing legal frameworks, and organizational readiness for transformation. This section analyzes the feasibility of the system while identifying the primary barriers that may arise during real-world adoption

7.3.1. Technical Feasibility: Standardization and Interoperability

From a technological perspective, VeriDocChain is built upon mature international standards established by the World Wide Web Consortium (W3C), including Decentralized Identifiers (DID) and Verifiable Credentials (VC). These standards provide a solid foundation for cross-border and cross-domain interoperability, enabling the system to integrate flexibly with existing digital services and emerging Web3 platforms.

However, the most significant technical challenge lies in connecting blockchain infrastructure with legacy information systems, including electronic health records (EHR), academic databases, and enterprise management systems. These legacy platforms are typically based on centralized architectures, heterogeneous data formats, and limited application programming interfaces (APIs), necessitating customized integration solutions and substantial deployment resources. In addition, scalability remains a critical concern, as the network must sustain stable transaction throughput when serving large-scale user populations.

7.3.2. Legal and Privacy Feasibility

VeriDocChain aligns closely with Vietnam's national digital transformation agenda and the current legal framework, particularly the 2023 Law on Electronic Transactions and Decree No. 13/2023/ND-CP on personal data protection. The system adopts an off-chain/on-chain hybrid storage model, in which sensitive identity data are stored off-chain, while only cryptographic hashes and digital signatures are anchored on the blockchain as verifiable proof.

This architectural approach effectively resolves the inherent tension between blockchain immutability and the "right to be forgotten" mandated by data protection regulations. It enables legally compliant deletion of original data upon request without compromising the integrity of the distributed ledger.

7.3.3. Organizational and Digital Transformation Challenges

Beyond technical and legal considerations, organizational readiness is a decisive factor in determining the success of the ecosystem. The transition from paper-based administrative processes to a decentralized digital credential management model requires a comprehensive change management strategy, encompassing workforce training, process reengineering, and cultural adaptation within organizations.

From a market perspective, the value of VeriDocChain is strongly influenced by network effects; the system reaches its full potential only when issuers, holders, and verifiers participate collectively. Resistance to change among traditional institutions, combined with the learning curve faced by end users in adopting concepts such as digital wallets and cryptographic key management, presents substantial challenges. These barriers must be addressed through structured pilot programs, continuous education, and strategic communication initiatives.

7.4. Risk Analysis for Use Cases

Successful deployment demands proactive identification and mitigation of technical and operational risks:

Table 3: Risk Assessment and Mitigation Strategies for VeriDocChain Use Cases

Risk	Impact Level	Mitigation Strategy
Private key loss	High	Social recovery protocols and secure backup infrastructure
Identity fraud during onboarding	High	Mandatory biometric eKYC and integration with national databases (e.g., FPT.IDCheck)
Regulatory changes	Medium	Adaptive compliance via updatable smart contracts and continuous legal monitoring
Supply chain data fraud	Medium	Blockchain-IoT integration, tamper-proof packaging, big data QR anomaly detection
Scalability bottlenecks	Medium	Multi-chain architecture and Layer-2 scalability solutions

7.5 Impact Summary Across Domains

Table 4: Quantitative Impact Assessment of VeriDocChain Across Application Domains

Sector	Time-Saving Indicators	Cost-Saving Indicators	Quality Impact
Education	Credential verification time reduced from 14 days to less than 1 minute.	80% reduction in operational costs for academic and registrar departments.	Complete elimination of diploma fraud (97% success rate in prevention).
Healthcare	Instant access to electronic medical records and health history across hospitals.	Significant reduction in redundant testing costs due to trusted shared data.	Enhanced security and privacy for sensitive patient information.
Supply Chain	Document processing time (LC, CO) reduced from 10 days to 4 days on average.	33-37% operational cost reduction by eliminating intermediaries.	Data accuracy improved by 38%, leading to a 25% drop in dispute management costs.
Real Estate	Contract closing time reduced from a standard 30-day period to just a few	50-90% reduction in transaction costs and legal fees through automation.	Asset liquidity increased by 52.1% through streamlined digital processes.

	days.		
Finance (KYC)	New customer onboarding speeds improved by 40-50%.	25-50% reduction in identity verification and regulatory compliance costs.	Creation of immutable audit trails for tax and banking regulatory inspections.

These figures confirm that VeriDocChain not only provides technological advantages but also generates significant and direct economic value for organizations and the entire society.

PART IV – BUSINESS MODEL & ECONOMIC DESIGN

8. Business Model Design

Building on the use cases that have been technically and procedurally validated in Sections 6 and 7, Section 8 focuses on transforming trust flows into sustainable revenue mechanisms. VeriDocChain is not designed merely as a conventional IT tool, but as a Trust Service Platform operating under a hybrid model that combines Blockchain-as-a-Service (BaaS) and Web3 Software-as-a-Service (SaaS). By reducing verification costs from several dollars to mere cents and shortening processing time from weeks to seconds, the platform generates significant economic surplus for society while simultaneously ensuring long-term profitability for investors.

8.1. Customer Segments

Unlike the operational role analysis presented in Section 7, customer segmentation from an economic perspective emphasizes willingness to pay and the cost-optimization incentives of each stakeholder group within specific use cases.

8.1.1. Issuers – Sources of Value Supply

This group includes universities, hospitals, professional associations, and government regulatory agencies—entities that are responsible for issuing credentials and bear legal accountability for the validity of digital certificates. In use cases related to degree issuance, medical certifications, and professional licenses (Sections 6.1–6.3), these organizations face significant pressure to protect institutional reputation, prevent fraud, and reduce administrative costs.

For issuing organizations (Issuers), VeriDocChain delivers economic value by reducing operational expenditure (OpEx) associated with manual verification and physical record storage through automated verification and digitized record management. In addition, the platform mitigates legal risks related to credential issuance and revocation through transparent blockchain-based mechanisms, while enhancing institutional credibility via instant and globally verifiable authentication.

8.1.2. Verifiers – Sources of Value Demand

This segment includes banks, insurance companies, employers, customs authorities, and credential-receiving organizations. In use cases involving financial KYC, recruitment, and supply chain customs clearance (Sections 6.2–6.4), this group represents the primary revenue-generating customer base, as these stakeholders are willing to pay for access to pre-verified data in order to reduce risk and accelerate decision-making processes.

The application of the Self-Sovereign Identity (SSI) model enables these organizations to reduce KYC and compliance costs by 25–50%, while shortening customer or employee onboarding time from several days to just a few minutes. As a result, the platform significantly minimizes credential fraud and data inconsistencies throughout the verification process.

8.1.3. Ecosystem Partners

This segment comprises providers of ERP, LMS, HRM, HIS, and other Web3 platforms. Although indirect customers, ecosystem partners play a strategic role in market expansion. By integrating VeriDocChain into existing systems, these partners enhance the value proposition of their own products and contribute to the creation of network effects across the broader ecosystem.

8.2. Revenue Streams and Pricing Strategy

VeriDocChain implements a hybrid pricing strategy designed to maximize customer lifetime value (LTV) and minimize adoption barriers during the early stages of market expansion.

Table 5: VeriDocChain Revenue Streams & Pricing Strategy

Revenue Model	Paying Entity	Fee Structure	Economic Characteristics	Linked Use Cases
Transaction-based Fee (by Use Case)	Issuer / Verifier	• Fee per issued Verifiable Credential (VC) (paid by Issuers) • Fee per verification request or query (paid by Verifiers)	Pay-as-you-go model, suitable for small organizations and the public sector; payments can be settled using the VDG utility token in a multi-chain environment	6.1. Education 6.2. Healthcare 6.3. Recruitment 6.4. Customs

SaaS Subscription Model	Enterprises, large organizations	Periodic fees (monthly/annual) based on usage scale and feature packages	Stable recurring revenue, optimized LTV; supports large-scale credential management	Recruitment enterprises Banks, insurance companies
White-label & Custom Integration (Enterprise / B2G)	Government, large corporations	Initial implementation fee plus annual maintenance and operational fees	High-value revenue streams, long-term contracts, aligned with B2G strategies	6.1.3. National-level notarization & authentication
VDG Utility Token	Entire ecosystem	Token used for transaction fee payments (Gas / Service Fees)	Supports the digital economy and expansion of the Web3 ecosystem	All Use Cases

8.3. Cost Structure

VeriDocChain operates under a Blockchain-as-a-Service (BaaS) model, enabling the conversion of upfront capital expenditure (CapEx) into flexible operational expenditure (OpEx). This approach optimizes marginal costs and supports early break-even as the platform scales. The main cost components include:

- **Development and Maintenance Costs (R&D):** Accounting for approximately 15–25% of total annual costs, these expenses focus on core system development, updates to W3C standards (DID, Verifiable Credentials), smart contract enhancements, and compliance with evolving regulatory frameworks such as Decree No. 13/2023/NĐ-CP and GDPR.
- **Infrastructure and Operational Costs:** Leveraging the TrustKeys Network reduces initial deployment time by approximately 60% and lowers total cost of ownership (TCO) by 30–50% compared to building a proprietary blockchain infrastructure.
- **Compliance and Legal Costs:** Audit processes and credential lifecycle management are automated through smart contracts, thereby reducing legal costs while enhancing transparency in compliance operations.
- **Customer Acquisition and Go-to-Market Costs (CAC):** These include expenses related to pilot deployments, user training, and organizational change management during the transition from paper-based systems to digital verification platforms.

8.4. Key Partnerships and Distribution Channels

The scalability and economic effectiveness of VeriDocChain are strengthened through **strategic partnerships and distribution channels** that support deployment, expand market reach, and increase platform adoption. The main partner groups and channels include:

- **Infrastructure and Technology Partners:** Including the TrustKeys Network, blockchain networks, and cloud service providers, which collectively support stable, secure, and scalable system operations.
- **B2B and B2G Distribution Channels:** Direct engagement with large enterprises, industry associations, and government agencies to promote platform adoption at organizational and institutional levels.

- **Academic Networks:** Universities and educational institutions act both as customers and as trust anchors, contributing to the credibility and legitimacy of digital credentials within the ecosystem.

These partnerships and distribution channels directly support VeriDocChain's go-to-market strategy and provide a solid foundation for its fundraising and scaling plans in subsequent phases.

PART V – GO-TO-MARKET STRATEGY

9. Market Opportunity and Competitive Landscape

The rapid growth of Vietnam's digital economy, supported by national digital transformation programs, is driving the shift in identity and credential verification from centralized systems to decentralized, internationally compliant solutions. VeriDocChain addresses this gap by combining the legal reliability and security of traditional systems with the flexibility and protection of blockchain, presenting significant market opportunities and competitive advantages.

9.1. Target Market Analysis

The target market of VeriDocChain is defined by customer groups with high demand for identity and credential verification, including educational institutions, recruitment enterprises, regulatory agencies, and individual users. The growing demand for identity and credential verification reflects the increasing urgency of establishing trust in an environment where electronic transactions are becoming ever more prevalent.

VeriDocChain's target market is divided into four main segments, each with distinct pain points and motivations for adopting blockchain-based verification solutions, thereby creating opportunities to optimize operational processes and enhance user experience across different customer groups.

9.1.1. Education and Training Segment

Education and training represent the most important beachhead segment for VeriDocChain. Vietnam currently possesses tens of millions of digitized educational records, yet verification processes remain heavily dependent on manual procedures or centralized databases. This results in high administrative costs, significant delays, and substantial risks of credential fraud.

Furthermore, the recognition of Vietnam's National Qualifications Framework (VQF) as equivalent to the ASEAN Qualifications Reference Framework (AQRF) has increased the demand for cross-border credential verification. Autonomous universities and international training institutions are the most highly motivated groups to adopt blockchain-based verification solutions in order to protect institutional reputation and standardize academic data.

Table 6: Indicators Demonstrating the Potential of the Vietnamese Education Market for Digital Credential Verification Solutions

Education Market Indicator	**Value/Data**	**Relevance to VeriDocChain**
Number of digitized education records	24.55 million records	Extremely large data volume requiring long-term verification demand
Growth of high school graduates	~10% annually	Continuous inflow of new data requiring VC issuance
Number of universities in QS Asia rankings	25 institutions (2026)	Institutions with strong incentives to protect academic reputation

Target for online public services in education	100% by 2030	Policy pressure driving digital verification

The indicators in Table 9.1 illustrate the large scale, stable growth, and increasing digital transformation pressure of Vietnam's education market, explaining why this segment is selected as the beachhead market in VeriDocChain's Go-to-Market strategy.

9.1.2. Recruitment and Human Resource Management Segment (HRTech)

Vietnam's HRTech market is rapidly expanding, with growing demand for recruitment and workforce management. Many enterprises continue to struggle with verifying the authenticity of credentials and candidate experience, resulting in significant consumption of time and resources.

VeriDocChain addresses this problem by delivering "instant trust." Instead of requiring 7 to 14 days for traditional background checks, enterprises can verify candidate qualifications in under one minute through Verifiable Credentials, thereby optimizing recruitment processes—particularly in high-demand sectors such as information technology.

9.1.3. Banking, Financial Services and Insurance Segment (BFSI)

Banks and financial institutions face increasing pressure to comply with KYC (Know Your Customer), AML (Anti-Money Laundering), and personal data protection regulations. The use of Self-Sovereign Identity (SSI) enables the reuse of verified information, reducing compliance costs and shortening customer onboarding time from days to minutes.

The development of Vietnam's digital asset trading market is generating strong demand for identity verification, with service providers required to perform strict KYC procedures. This creates a significant opportunity for VeriDocChain to provide identity verification infrastructure for exchanges and financial institutions during the pilot period.

9.1.4. Logistics and Supply Chain Segment

Vietnamese logistics enterprises are actively seeking blockchain solutions to enhance transparency. VeriDocChain can extend into verifying certificates of origin (C/O), quality certificates, and commercial documentation. Full digitization of such documents can reduce customs clearance time by approximately 30% and significantly lower operational costs by eliminating unnecessary intermediaries.

9.2. Competitive Positioning

The competitive landscape of VeriDocChain in Vietnam consists of domestic technology corporations, international startups, and state-developed infrastructure solutions.

9.2.1. Domestic Competitors: Traditional PKI Solutions

Companies like FPT IS and MISA are taking a prominent position in the field of digital signatures and electronic contracts. Their solutions have been widely deployed in banks and businesses, and are deeply integrated into management and accounting ecosystems, particularly serving the small and medium-sized enterprise (SME) segment.

However, these solutions are primarily based on traditional public key infrastructure (PKI) models and centralized certificate authorities (CAs). While possessing high legal validity, they often face limitations regarding:

- **Privacy:** Users often have to share entire documents instead of just the necessary information (Selective Disclosure).
- **Cost:** Annual registration fees and fees per digital certificate can be barriers to the widespread adoption of micro-credentials.
- **Flexibility:** Reliance on hardware devices like USB tokens is becoming obsolete compared to remote signing and digital identity wallets.

VeriDocChain is shifting from an "organization-based authentication" model to an "SSI-based self-identification" model. Instead of directly competing in providing digital signatures to businesses, VeriDocChain focuses on creating a globally interoperable certificate and qualification authentication layer, something that domestic PKI systems have not yet optimized.

9.2.2. International Competitors: Global Standards Providers

Solutions such as Microsoft Entra, Dock.io, and IBM Verify are shaping global standards for Verifiable Credentials and Decentralized Identifiers. Microsoft Entra, benefiting from integration with Azure and LinkedIn, is a particularly strong competitor in workplace and professional identity verification.

These competitors' advantages lie in their strict adherence to W3C standards and global scalability. However, their weaknesses in the Vietnamese market are:

- **Domestic compliance:** Regulations on data storage in Vietnam (Decree 13) and requirements for integration with national identification systems (VNeID) are technical and legal barriers for foreign providers.
- **Language and support:** The lack of on-site support and a localized Vietnamese language interface is a disadvantage in accessing government agencies and public universities.

9.2.3. Comparative Competitive Advantages of VeriDocChain

VeriDocChain differentiates itself through a “Hybrid Trust” model. By integrating with the TrustKeys platform, VeriDocChain not only delivers decentralized technical architecture but also ensures that all system identities undergo legally compliant eKYC under Vietnamese law.

Table 7: Comparison of Identity Verification and Trust Management Models

Criteria	**VeriDocChain**	**Domestic PKI Solutions (FPT/MISA)**	**International SSI Solutions**
Architecture	Decentralized (SSI)	Centralized (CA/PKI)	Decentralized (SSI)
Data control	User-held wallet	Organization-managed	User-held wallet
Privacy	Selective disclosure (ZKP)	Full document sharing	Selective disclosure (ZKP)
Interoperability	Global (W3C standards)	Closed ecosystem	Global (W3C standards)
Compliance with VN law	Optimized for Decree 13	Very high	Medium/Low
Operating cost	Low (Pay-as-you-go)	High (annual subscription)	Medium (enterprise licensing)

VeriDocChain's core competitive advantage lies in resolving the tension between blockchain immutability and the "right to be forgotten" under GDPR and Decree 13. Its hybrid on-chain/off-chain data strategy allows personal data deletion without compromising cryptographic integrity, making regulatory approval in Vietnam achievable.

10. Go-to-Market Strategy

VeriDocChain's GTM strategy goes beyond selling software; it focuses on building a new trust infrastructure. Success requires a multi-layered market entry roadmap combining demand creation from end users (pull strategy) with strategic partnerships among credential issuers (push strategy).

10.1. Market Entry Strategy

VeriDocChain adopts a phased market entry approach to minimize risk while maximizing early adoption.

- **Phase 1: Establishing the Beachhead (Years 1–2)**

In this phase, VeriDocChain focuses on the higher education and skills training segment to establish initial trust anchors for the ecosystem. The objective is to become the technology partner of at least five leading universities (including international institutions such as RMIT and BUV, as well as autonomous public universities such as Hanoi University of Science and Technology and the National Economics University), together with approximately ten reputable technology training centers.

To achieve this objective, the project deploys free or low-cost pilot programs (PoC), focusing on the issuance of diplomas and course certificates in the form of Verifiable Credentials. This approach helps build an initial user base of students and graduates, who will continue carrying their VeriDocChain identity wallets as they enter the labor market.

In parallel, VeriDocChain leverages national blockchain development policies, particularly Decision No. 1236/QD-TTg, to participate in key research and training programs, thereby strengthening the platform's legitimacy and adoption.

- **Phase 2: Ecosystem Expansion (Years 2–4)**

After establishing the initial user base and building a trusted on-chain credential repository, VeriDocChain shifts its focus toward expanding the ecosystem, targeting verifiers as the primary growth drivers. The goal of this phase is to integrate VeriDocChain's verification tools directly into the operational workflows of large corporations, recruitment enterprises, financial institutions, and HR platforms.

To achieve this, VeriDocChain deploys verification APIs under a “try-before-scale” model, allowing HR departments and enterprise partners to test the system at minimal or no cost during the initial stage. At the same time, the platform expands into the healthcare sector (verifying professional medical licenses) and the logistics sector (verifying certificates of origin and supply chain documentation), reinforcing VeriDocChain’s role as a shared verification infrastructure rather than a solution limited to the education sector.

At the regional level, VeriDocChain leverages ASEAN cooperation frameworks, particularly the AQRF, to implement cross-border credential verification pilot projects, with the objective of supporting Vietnamese workers seeking employment opportunities in Singapore, Thailand, and Malaysia.

- **Phase 3: Becoming National Infrastructure (Years 4–6)**

In its maturity stage, VeriDocChain positions itself not merely as an enterprise technology solution but as a national digital trust infrastructure for the public sector. The strategic objective of this phase is to become a core component of the National Blockchain Platform (NDAChain) or to be directly integrated into national digital identity systems such as VNeID.

This large-scale transition toward a B2G (Business-to-Government) model enables VeriDocChain to support the government in managing the national credential repository, providing digital identity services for citizens, and administering critical administrative certificates.

During this phase, VeriDocChain’s role evolves from a service provider into an infrastructure partner, accompanying government agencies in building digital government and the digital economy. National-level integration not only standardizes credential and identity verification mechanisms but also establishes the foundation for transparent and secure data interoperability across ministries, sectors, and local authorities.

10.2. Customer Acquisition and Growth Plan

VeriDocChain's growth strategy is built on network effects, in which the platform's value simultaneously increases for both issuers and verifiers as the number of ecosystem participants grows. Credential holders play a central role as the bridge driving verification demand from the labor market and user organizations.

10.2.1. Strategy for Issuers (Educational Institutions and Organizations)

For issuing institutions, the primary motivation for joining the platform is operational cost reduction and the protection of academic reputation. VeriDocChain applies a freemium model for the education sector, offering free initial setup and system maintenance while charging only a small fee per diploma or certificate issued as a Verifiable Credential. This approach enables institutions to shift upfront capital expenditures (CapEx) into flexible operational expenditures (OpEx), aligning with the budgetary characteristics of public and educational organizations.

In parallel, the platform provides digitization tools that assist institutions in converting legacy academic records from paper format into Verifiable Credentials, significantly reducing administrative workload and operational costs.

10.2.2. Strategy for Verifiers (Enterprises and Banks)

For verifiers, the core value delivered by VeriDocChain is speed and accuracy in information validation. The solution is designed for deep integration into recruitment and candidate evaluation processes. Partnerships with platforms such as TopCV enable the display of a "Verified by VeriDocChain" badge on candidate profiles. Enterprises using TopCV can perform credential verification with a single click at minimal cost.

In the banking and financial sector, VeriDocChain offers verification services under a SaaS model, supporting institutions in performing KYC and rapidly validating customer identity and educational background during account opening and credit assessment. Reducing onboarding time from three days to five minutes not only improves user experience but also delivers clear economic benefits for organizations with large transaction volumes.

10.2.3. Strategy for Holders (Individuals)

For individuals, the primary motivation for adopting the platform is convenience and career opportunity. VeriDocChain develops a multifunctional SSI identity wallet that enables users to store and manage multiple types of credentials, including micro-credentials, medical test results, and event tickets.

In addition, the platform implements communication campaigns centered on the concept of “data sovereignty,” helping users better understand their rights over personal data under existing legal regulations. When learners proactively request Verifiable Credentials instead of paper certificates, they simultaneously stimulate verification demand from employers, thereby strengthening the network effects of the ecosystem.

The table below summarizes the core value that VeriDocChain brings to each participating group in the ecosystem, while also clarifying the revenue generation mechanism and the maintenance of the platform's network effect.

Table 8: Value Proposition and Revenue Model by Stakeholder Group

Stakeholder	Core Value Delivered	Revenue Model
Educational Institutions / Organizations	Fraud prevention, 90% reduction in verification time	Pay-per-issue VC fees

Enterprises / HR	Accurate hiring, 50% reduction in background check cost	Pay-per-query or SaaS subscription
Individuals	Full ownership of credentials, 24/7 job application	Free (user base generation)
Government Agencies	Transparency, support for digital government	White-label deployment & system maintenance fees

10.3. Partnership and Ecosystem Development

The success of VeriDocChain depends on its ability to establish strategic partnerships that integrate the platform into existing infrastructure and ecosystems, thereby accelerating market entry and enabling scalable growth.

10.3.1. Infrastructure and Digital Identity Partner: TrustKeys

TrustKeys serves as VeriDocChain's strategic infrastructure partner. Integration with TrustKeys' validator node network enables decentralization without heavy upfront physical infrastructure investment. Furthermore, the use of TrustKeys' identity modules ensures compliance with Vietnamese regulations on digital signatures, PKI, and electronic identity, enhancing platform legitimacy in regulated sectors such as education and public administration.

10.3.2. HRTech and Career Platform Partners

In recruitment and human resource management, VeriDocChain establishes strategic partnerships to become the default verification provider on the TopCV platform—currently the most effective distribution channel for reaching verifiers. In parallel, VeriDocChain utilizes the "Verified on LinkedIn" APIs to allow users to display verified credentials directly on global LinkedIn profiles, significantly increasing the international utility of Verifiable Credentials.

10.3.3. Government and Association Partners (B2G)

VeriDocChain pursues strategic partnerships with regulatory authorities and industry organizations to promote the formation and dissemination of national digital credential verification standards in Vietnam. The Ministry of Education and Training (MOET) plays a central role through participation in technical standard drafting committees for national digital credentials, creating conditions for VeriDocChain's standards to be gradually recognized and adopted across the education system.

In addition, cooperation with the Vietnam Blockchain Association (VBA) and the Vietnam Software and IT Services Association (VINASA) enables VeriDocChain to leverage extensive networks of enterprises, experts, and media organizations to conduct outreach, policy advocacy, and blockchain workforce development, thereby fostering the sustainable growth of Vietnam's digital verification ecosystem.

10.3.4. International Technology and Training Partners

At the international level, VeriDocChain aims to collaborate with organizations such as CompTIA, Tether, and RMIT to deploy digital skills training programs and issue certifications directly on the VeriDocChain platform. This initiative creates a high-quality and immediately applicable supply of Verifiable Credentials for the IT workforce.

PART VI – FEASIBILITY, RISK & FUNDRAISING

11. Feasibility Risk Assessment

The feasibility assessment of VeriDocChain is based on a combination of empirical evidence obtained from the technical deployment process and an analysis of external environmental factors, aiming to determine the system's readiness for practical deployment as well as the associated strategic risks.

11.1. Technical Feasibility

11.1.1. Statement of Feasibility

The VeriDocChain system has demonstrated comprehensive technical feasibility at the proof-of-concept level. Experimental results show that the system architecture has gone beyond the realm of theoretical ideas, achieving stable operation of fundamental functions, including certificate registration, decentralized authentication, and the enforcement of data immutability directly on the blockchain environment. The success of this phase lays a solid foundation for integrating the system into larger-scale trust infrastructures such as the TrustKeys network in subsequent development phases.

11.1.2 Evidence from Implementation

Based on the technical implementation process, the system's feasibility is reinforced through the following four key empirical points:

The smart contract architecture operates precisely: The DocumentRegistry contract has been optimally designed to manage the mapping between the document's hash and the issuer's wallet address and timestamp. This mechanism establishes an immutable data "anchor," allowing for verification of the source of truth without directly storing sensitive data on the chain.

Figure 10: Core structure of DocumentRegistry smart contract

Successful deployment on Blockchain infrastructure: The source code has been successfully compiled and initialized on the Remix VM environment (London). The successful creation of unique contract addresses and the execution of transactions without system errors have confirmed the solution's compatibility with current blockchain standards.

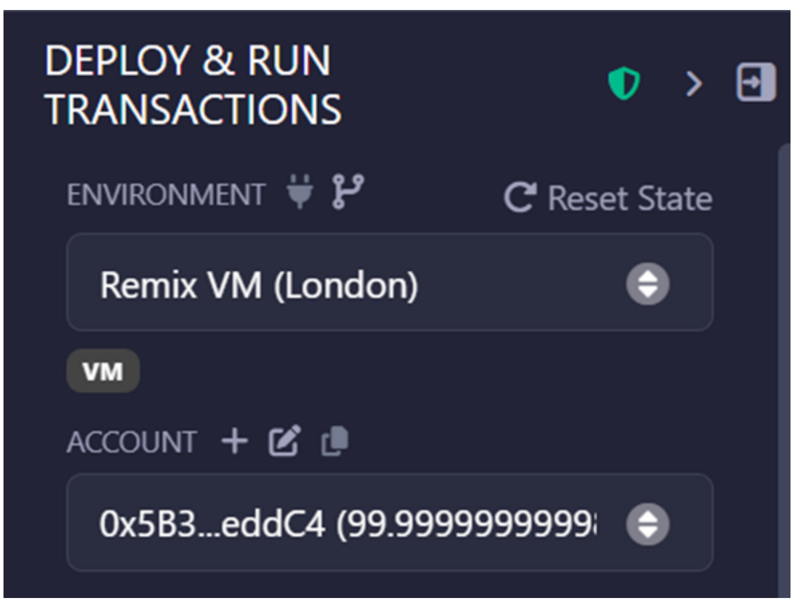

Figure 11: Deployment and execution environment configuration using Remix VM (London) for smart contract testing

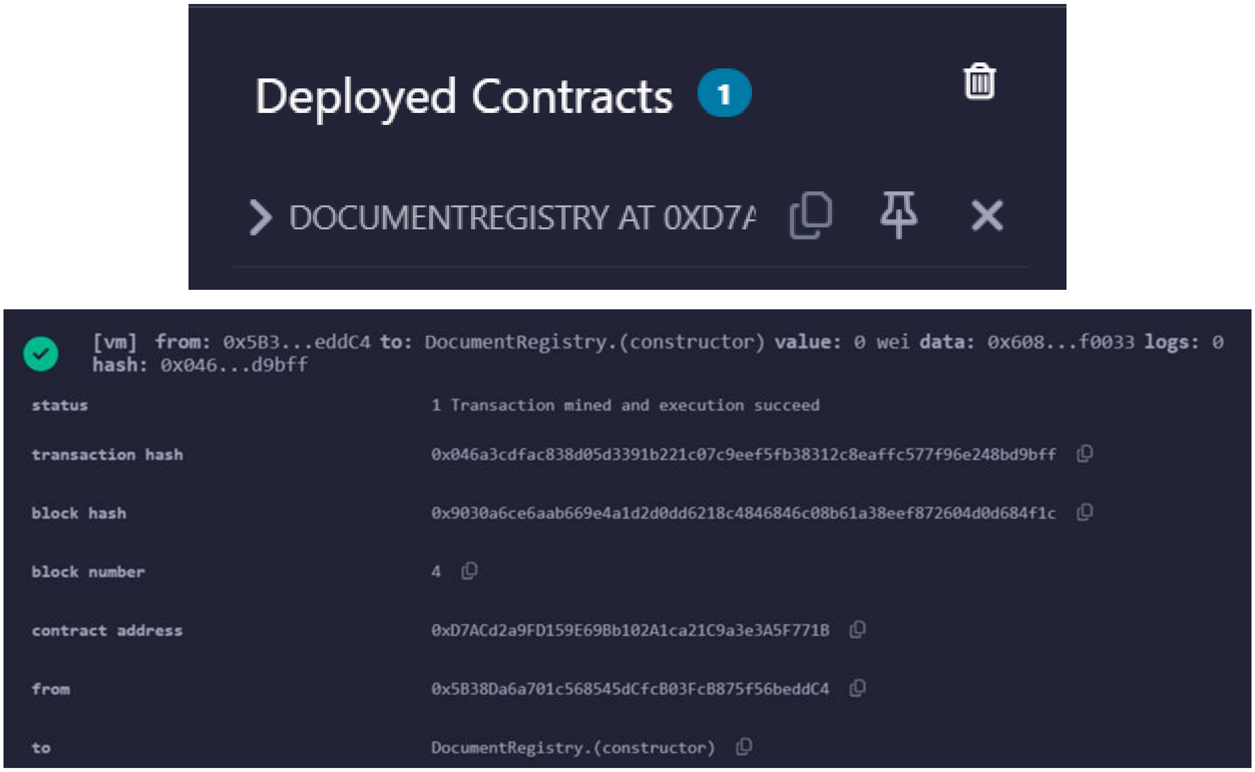

Figure 12: Successful deployment of smart contract on Remix VM

Stable operation of business functions: The system successfully implemented the document registration process via the registerDocument function and instant verification queries via the verify function. The returned results allow any third-party authenticator to check the authenticity of the document in real time with minimal latency.

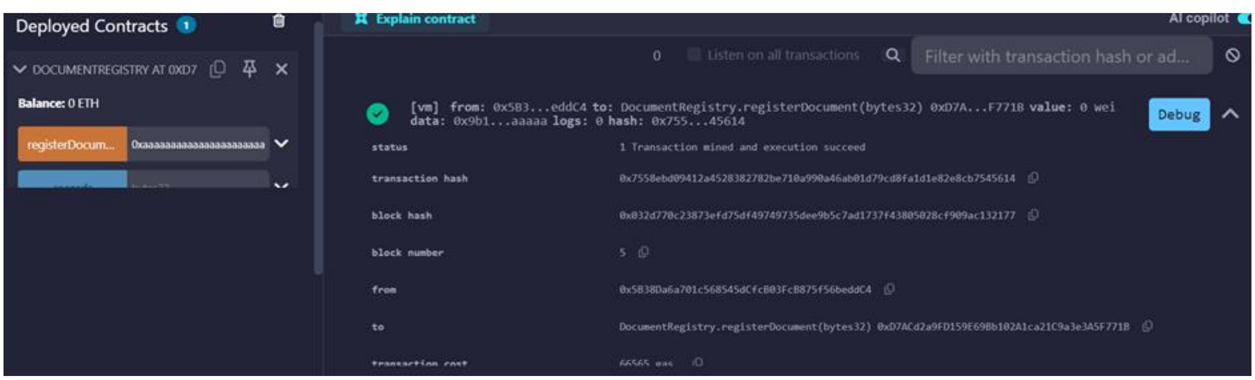

Figure 13: On-chain registration of a document hash using the registerDocument() function

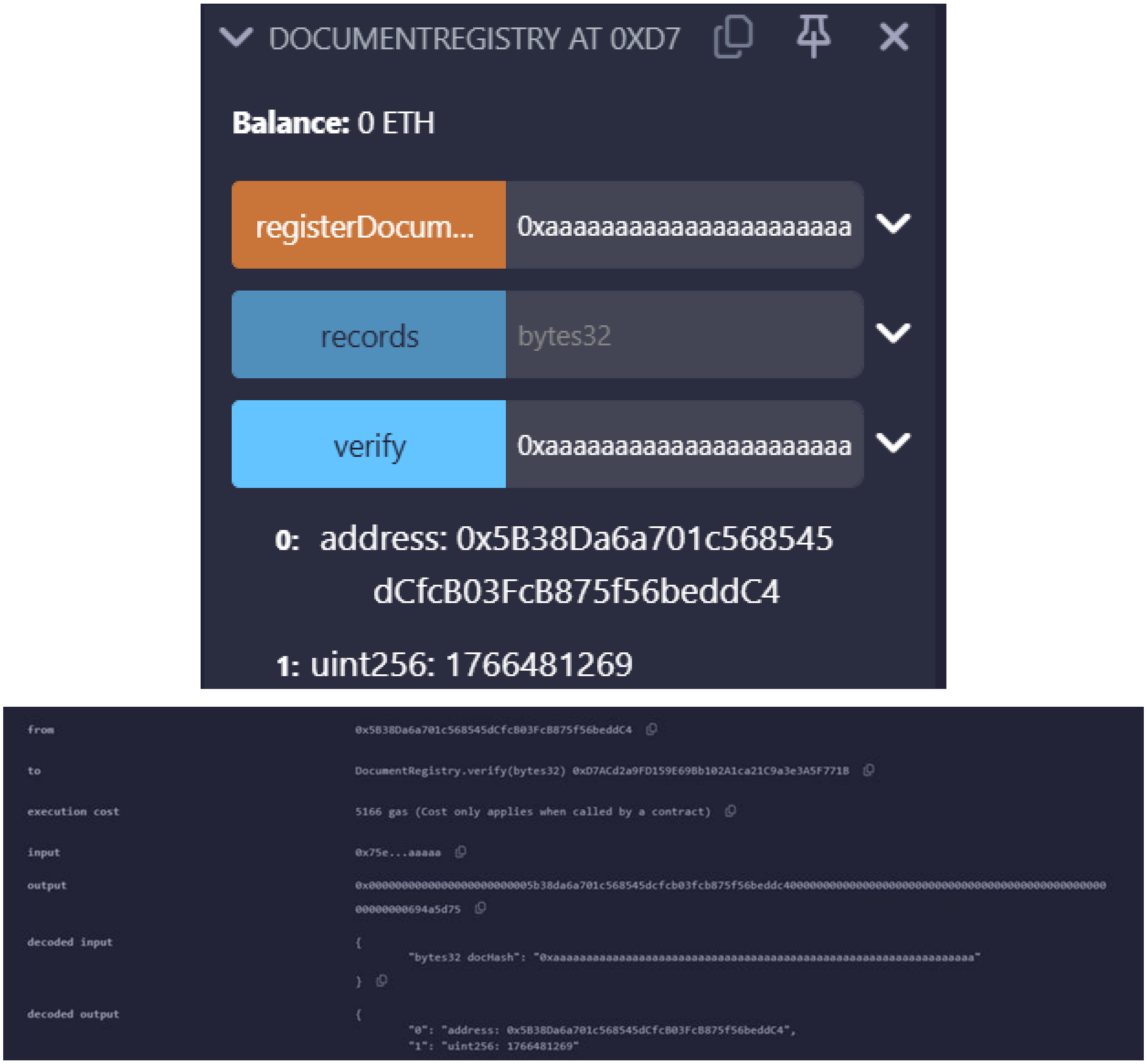

Figure 14: Result of on-chain document verification

Integrity and immutability are absolutely guaranteed: Experiments have successfully verified the mechanism for preventing duplicate registrations. When attempting to overwrite an existing hash code, the transaction is immediately reversed, confirming the fraud resistance and high security of the certificate management system.

Figure 15: Reverted transaction demonstrating immutability enforcement

11.2. Legal and Regulatory Feasibility

The legal feasibility of VeriDocChain is reinforced by the rapid maturation of Vietnam's regulatory framework governing electronic transactions and data governance during the national digital transformation period. The system not only complies with existing regulations but is also designed to leverage newly established legal mechanisms in order to ensure that digital diplomas possess legal validity equivalent to traditional paper documents.

VeriDocChain is architected in accordance with the Trust Services framework stipulated in Decree No. 23/2025/ND-CP (effective from April 2025). The system employs specialized electronic signatures to authenticate the identity of signatories and timestamping services to guarantee the integrity and existence time of issued credentials. The digital signature certificates supporting these trust services are valid for up to five years, thereby contributing to the long-term stability of the national digital certificate infrastructure.

To address the inherent conflict between blockchain immutability and privacy protection requirements, the system implements a Hybrid Storage architecture. Only cryptographic proofs (hashes) and revocation status are recorded on-chain, while personally identifiable information (PII) is securely stored in encrypted off-chain data repositories. This design enables the enforcement of the "right to be forgotten" by allowing original data to be erased off-chain without compromising the integrity of the distributed ledger.

Furthermore, compliance with W3C standards for Decentralized Identifiers (DID) and Verifiable Credentials ensures that credentials issued by VeriDocChain are interoperable with the ASEAN Qualifications Reference Framework (AQRF), thereby facilitating cross-border credential recognition without the need for complex legal legalization procedures. Collectively, these factors demonstrate that the system not only satisfies current legal requirements but also aligns with long-term trends in digital governance and electronic transactions.

11.3. Operational Feasibility

The operational feasibility of VeriDocChain is reflected in the level of readiness and adaptability of stakeholders, as well as in the data governance processes within the education and labor ecosystems.

The widespread adoption of national digital identity platforms such as VNeID in Vietnam has established a favorable behavioral foundation for the Self-Sovereign Identity (SSI) model. Building upon this foundation, VeriDocChain designs an interaction workflow consisting of four continuous stages: identity onboarding, issuance of digital credentials in the form of Verifiable Credentials, presentation of credentials with selective disclosure mechanisms, and instant verification. This approach significantly reduces friction in the user experience while simultaneously enhancing the overall trustworthiness of the entire process.

According to the national roadmap, biometric authentication will become a mandatory requirement for digital identity and electronic wallets. The integration of strong authentication methods, including facial recognition and fingerprint verification, enables VeriDocChain to eliminate identity fraud risks from the onboarding stage, ensuring that each decentralized identity (DID) is securely linked to a real individual.

In traditional systems, the revocation of fraudulent or erroneous credentials is extremely difficult. VeriDocChain addresses this challenge through an on-chain Revocation Registry. When a credential is revoked, the issuing authority only needs to update its status on the smart contract, after which the system automatically broadcasts the "Invalid" status to all verifiers in real time.

From an integration perspective, leveraging Blockchain-as-a-Service platforms such as TrustKeys Network allows universities and educational institutions to connect their existing management systems with VeriDocChain via APIs and SDKs, thereby significantly reducing infrastructure costs and administrative burdens.

11.4. Financial Feasibility

The financial feasibility assessment of VeriDocChain is based on the Web3 SaaS economic model, its capability to optimize operational costs, and the support provided by national-level digital transformation strategies.

Compared to traditional manual processes that are costly and time-consuming—requiring 2 to 4 weeks to verify a single academic credential—VeriDocChain reduces the verification time to under one minute with a very low marginal cost. This efficiency enables organizations to significantly reduce administrative expenses related to examination management and credential storage.

Furthermore, the domestic policy environment is creating favorable conditions for core technology and digital transformation projects to access government funding sources as well as B2G cooperation mechanisms. These conditions substantially reduce financial barriers during the scaling and expansion phase.

From a long-term operational perspective, the SaaS model allows VeriDocChain to maintain high profit margins by optimizing multi-chain transaction costs and developing a stable customer base with low churn rates. Collectively, these factors indicate that the system's financial model has the potential to achieve an early break-even point and sustain long-term growth.

12. Fundraising and Financial Roadmap

Based on the business model analyzed in **Part 8**, the financial roadmap of VeriDocChain focuses on transforming investment capital into a scalable trust infrastructure, leveraging the network effects of the self-sovereign identity (SSI) ecosystem. This approach aims to ensure a balanced combination of long-term growth, risk management, and financial sustainability in a context where regulatory frameworks and market acceptance are still evolving.

12.1. Funding Requirements and Capital Allocation

During the 2025–2026 period, VeriDocChain plans to conduct two strategic fundraising rounds to finalize product development and establish an initial market position in the education and credential verification sector in Vietnam.

The **Seed round** is expected to raise between **USD 500,000 and USD 1.5 million**, with the primary objectives of completing the Minimum Viable Product (MVP), deploying pilot projects with selected leading universities, and building early brand recognition within the education and HRTech ecosystem.

After achieving **Product–Market Fit**, the project targets a **Series A round** with an anticipated scale of **USD 2–8 million**, depending on user growth rate and market adoption. This capital will be allocated to expanding into high-verification-demand sectors such as healthcare and logistics, while progressively entering the ASEAN regional market.

VeriDocChain's capital allocation strategy follows the risk governance principles of Web3 infrastructure startups, with the following projected structure:

- **Core Development and R&D (30%)**: Enhancing smart contracts, integrating Zero-Knowledge Proofs (ZKP) to strengthen privacy, and ensuring compatibility with the latest W3C standards.

- **Operations and Infrastructure (20%)**: Maintaining node operations on the TrustKeys Network, optimizing off-chain storage, and ensuring high system availability.

- **Marketing and Ecosystem Development (25%)**: Attracting end users (students) and building employer communities through platforms such as TopCV.

- **Legal and Compliance (10%)**: Ensuring continuous compliance with personal data protection regulations (Decree 13) and preparing licensing documentation for trust services under Decree 23.

- **Risk Contingency (15%)**: Managing multi-chain transaction fee volatility and emergency security situations.

This allocation structure reflects the characteristics of blockchain infrastructure projects, in which technology R&D and regulatory compliance typically account for a higher proportion of total expenditure than in traditional SaaS business models.

12.2. Revenue Projection and Sustainability Model

VeriDocChain's financial model is designed to maintain high gross profit margins typical of SaaS platforms, while prioritizing long-term sustainability and scalability. The primary revenue streams include recurring subscription income from enterprises and financial institutions utilizing verification services via API, as well as transaction-based revenue from the issuance and verification of Verifiable Credentials by educational institutions under a pay-per-use model.

Beyond the private sector, the project holds strong potential to access digital transformation programs and public services through B2G partnerships, generating stable medium-term revenue. VeriDocChain's financial sustainability is supported by optimizing the ratio between Customer Lifetime Value (LTV) and Customer Acquisition Cost (CAC). With the support of the TrustKeys network, the project's CAC is projected to be 30–40%

lower than that of traditional competitors due to the availability of a pre-verified identity user base.

12.3. Growth and Scaling Roadmap

The growth roadmap is designed to transform VeriDocChain from a niche solution into a national trust infrastructure:

- **Phase 1 (Years 1–2):** The project focuses on establishing initial "trust anchors" in the education sector through partnerships with selected universities and reputable skills training centers. The objective of this phase is to validate the digital credential issuance and verification model and build the initial user dataset that forms the foundation of network effects (e.g., becoming a partner of at least five top Vietnamese universities and issuing Verifiable Credentials to 100,000 graduates).

- **Phase 2 (Years 2–4):** VeriDocChain expands the ecosystem toward verifiers by deeply integrating into recruitment workflows of major corporations and banks. In parallel, the project gradually pilots cross-border credential verification scenarios within ASEAN cooperation frameworks, thereby expanding platform usage and interoperability.

- **Phase 3 (Years 4–6):** VeriDocChain aims to position itself as a national-level trust infrastructure through B2G partnerships. The long-term objective is to integrate digital diplomas and certificates into national identity platforms (such as VNeID), supporting large-scale education and workforce data management while contributing to national blockchain development strategies.

The quantitative objectives within this growth roadmap are strategic benchmarks and may be flexibly adjusted according to market conditions, regulatory progress, and stakeholder adoption levels.

PART VII – CONCLUSION & FUTURE OUTLOOK

Through a comprehensive analysis of the system architecture, business processes, and core technical mechanisms, this study has clarified how the VeriDocChain system leverages blockchain technology and a self-sovereign identification (SSI) model to address the problem of digital diploma authentication in a transparent, reliable, and decentralized manner. The system architecture is designed to clearly separate the roles of Issuer, Holder, and Verifier, thereby reducing reliance on centralized intermediaries while enhancing learners' control over their academic credentials.

Beyond conceptual design, the study further demonstrates the technical feasibility of the proposed solution through the implementation and validation of a core smart contract. The deployment and testing results confirm that VeriDocChain can effectively support on-chain document anchoring, decentralized verification, and immutability enforcement without storing sensitive academic data on the blockchain. This technical validation strengthens the credibility of the proposed architecture and confirms that the system design can be translated into a functioning blockchain-based solution using open standards and existing development environments.

Despite these strengths, the study also acknowledges several challenges that may affect real-world adoption, including operational costs, scalability considerations, and acceptance within existing administrative and institutional environments. Nevertheless, the results indicate that VeriDocChain has the potential to serve as a reference model for digital diploma authentication systems within the broader context of digital transformation in education. More importantly, the system demonstrates the practical applicability of W3C standards and decentralized technologies in building sustainable and interoperable credential verification infrastructures.

In future development stages, the VeriDocChain system can be further enhanced both technically and in terms of its deployment model. From a technical perspective, integrating advanced cryptographic techniques such as Zero-Knowledge Proofs could strengthen privacy preservation while maintaining verifiability. In addition, potential integration with national digital infrastructure platforms and trusted ecosystems would expand the system’s applicability and institutional acceptance. At the same time, optimizing operational mechanisms and establishing a sustainable cost model will be critical for supporting large-scale adoption. Overall, these results demonstrate that VeriDocChain is not merely an academic exploration, but a technically validated and forward-looking solution with strong potential to evolve into a practical digital diploma authentication platform capable of meeting long-term educational and administrative requirements.

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